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Enhancing label distribution generation for lightweight facial expression recognition with local-global features and transformer guidance

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Thesis approval certificate goes in here.

I dedicate this work to my mother, who gave me this life and education, and to my father, who watches over me from the next one and inspired me to be an academic and professional.

Dedicado a mi madre, quien me dio esta vida y educación, y a mi padre, quien me observa desde la siguiente y me inspiró a ser académico y profesional.

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Abstract

Lightweight facial expression recognition (FER) models are essential for resource-constrained deployment, yet they typically lag behind heavier transformer-based approaches in accuracy. This thesis investigates whether enhancing the label distribution generator (LDG) used to train EfficientFace, a 1.28-million-parameter lightweight FER network, can close that accuracy gap without increasing the deployed model’s cost. Three LDG enhancements are evaluated: FER-pretrained APViT backbone weights, EfficientFace local-global feature modules, and label distribution learning guided by APViT outputs. The study follows a two-phase factorial design on the AffectNet dataset (7- and 8-class subsets).

Phase I ablates the three enhancements on the modified LDG (M-LDG). APViT transformer guidance proved the only statistically significant factor, lifting M-LDG accuracy by 7.889 percentage points to 64.34% on AffectNet-7 and 57.71% on AffectNet-8. FER-pretrained weights and local-global modules contributed negligibly, due to mismatch with the M-LDG’s deeper ResNet-50 backbone. Phase II trains two EfficientFace variants, one guided by the best M-LDG and one directly by APViT, and compares them against the published EfficientFace baseline.

The main hypothesis is rejected: neither variant surpassed the published baseline of 63.70% on AffectNet-7 and 59.89% on AffectNet-8. The best result, M-LDG-guided EfficientFace, reached 63.00% and 55.99% on the two subsets. A capacity bottleneck is identified: a parameter ratio of approximately 44:1 between M-LDG and EfficientFace is too wide for label distribution learning to bridge, unlike the modest gap in the original EfficientFace paper. Replacing the M-LDG with APViT as the direct teacher produced no statistically significant difference in student accuracy, suggesting that at

this capacity scale the choice of teacher is secondary to the structural gap. Future work should address the bottleneck directly, for example through intermediate-capacity assistant networks or distribution sharpening techniques.

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Glossary

affective computing A field of research concerned with systems that detect, understand, or replicate human emotions; facial expression recognition is one of its application areas.

AffectNet Large-scale in-the-wild facial expression dataset (approximately 1 million images) with 7 and 8-class subsets, widely used for training and evaluating FER models.

discrete emotion model A theory that represents emotions as a fixed set of categories (e.g., the six basic emotions: anger, disgust, fear, happiness, sadness, surprise), often extended with neutral and contempt.

edge computing Computing performed near the source of data (e.g., on devices or local servers) rather than in a central cloud, often requiring efficient models for real-time or offline use.

EfficientFace A lightweight facial expression recognition network proposed by Zhao et al., built on ShuffleNet V2 with local-global feature extraction modules and

trained via label distribution learning using a label distribution generator.

emotion distribution A vector of description degrees (intensity values per emotion class), one per emotion class, that sum to one; used in emotion distribution learning to represent the intensity of each emotion in a facial expression.

facial expression recognition A field within affective computing that seeks to detect human emotions based on facial expressions; FER systems provide automated solutions that analyze facial features to identify the emotion being conveyed, typically through a pipeline of preprocessing, feature extraction, and classification.

in-the-wild (ITW) Referring to data or conditions that are not controlled in a laboratory: natural variation in pose, occlusion, illumination, and context; FER ITW denotes facial expression recognition on such unconstrained images or datasets.

knowledge distillation Training a smaller or more efficient model to mimic the outputs of a larger or more accurate model, so that the former learns from the latter's soft predictions.

label distribution generator A network that produces soft label (emotion) distributions for training images; typically trained first and then used to generate targets for training a lightweight network, and can be discarded at inference.

label distribution learning A learning paradigm where the target is a distribution over labels (e.g., description degrees per class) rather than a single label; a lightweight network may learn from soft distributions produced by a stronger model or a label distribution generator.

lightweight FER Facial expression recognition approaches designed for resource-constrained settings (e.g., mobile or edge devices), balancing accuracy with low parameter count and computational cost.

local-global feature extraction An architectural strategy that captures both local details (e.g., fine-grained facial regions) and global context (e.g., overall expression).

multi-head self-attention An extension of self-attention that linearly projects the query, key, and value vectors once per attention head, computes self-attention for all heads in parallel, and combines the results; proposed by Vaswani et al. to learn information at different representation levels, and used in stacked layers of the Transformer encoder and decoder.

patch embedding In Vision Transformers, the representation obtained by splitting an image into patches, flattening them, and linearly projecting to a sequence of vectors used as input to the transformer encoder.

residual connection A skip connection that adds the input of a block to its output, enabling training of very deep networks by easing gradient flow; used e.g. in ResNets.

RetinaFace A face detection and alignment method that preprocesses facial images, ensuring consistent pose and landmark alignment.

self-attention A mechanism that computes a representation of a sequence by relating each position to every other position via query-key-value operations; often refers to scaled dot-product attention as proposed by Vaswani et al., used in Transformers and Vision Transformers.

static FER Facial expression recognition on single images using only spatial information, as opposed to dynamic FER which uses video or image sequences.

Transformer A neural network architecture based on stacked self-attention layers, originally proposed for sequence transduction; uses an encoder-decoder structure, with the encoder used in Vision Transformers for image classification.

Chapter 1

Introduction

“The face is a picture of the mind
with the eyes as its interpreter.”
Marcus Tullius Cicero

Facial expression recognition (FER) plays a crucial role within human-computer interaction (HCI), enabling automated detection of human emotions from facial data [20]. FER has also attracted researchers in the field of affective computing, due to its potential applications in creating systems that detect, understand, and replicate emotions [16]. Thanks to advancements in computer vision and deep learning (DL) methods, FER technology has improved significantly over the recent years [20]. This thesis focuses on the subfield of lightweight FER for deployment on resource-constrained environments, which demands a balance between accuracy and efficiency.

FER systems have a wide range of applications, including sociable robots, medical treatment, driver safety, marketing, education, psychological research, and security [2; 16; 20; 31]. Despite this, FER faces several challenges such as occlusion, pose and lighting conditions, subject variability, data annotation, resolution, among others [20; 31]. On top of this, real-world usage scenarios such as edge computing require the highest level of efficiency in order to be feasible [34].

Given the proliferation of mobile and edge platforms with a limited resource budget, there is an increasing need for lightweight FER models. Recent studies have focused on creating optimized deep learning methods for computer vision [14; 38], with some even applying such methods directly to FER [29; 40]. These studies have shown that it is possible to significantly optimize these models while maintaining or even improving the system’s accuracy.

This thesis aims to enhance the lightweight FER network, EfficientFace [40], in order to improve its accuracy while maintaining its low resource cost. To this end, we modify the authors’ training approach by leveraging the APViT [36] model, which achieves

state-of-the-art results in FER.

1.1. Problem

Facial expression recognition (FER) systems provide automated solutions that analyze facial features to identify the emotion being conveyed. In the context of static or image-based FER, feature-extracting CNNs followed by fully-connected classifiers have been the generally predominant approach [2]. Following the wider trend in deep computer vision, networks with self-attention mechanisms have been applied to the FER task to obtain significant classification improvements [15; 23; 29; 30; 35; 36]. In particular, Vision Transformers are quite adept at the FER task when training with challenging pose, lighting and occlusion conditions (in-the-wild). This is, in part, thanks to the network’s ability to pay greater attention to more important facial regions, which mitigates the effect of occlusion and pose variations [16].

While several of these approaches have achieved state-of-the-art results, many of them incur a heavy resource cost in terms of network parameters and floating point operations per second (FLOPs). When considering that the feature extraction step is the system’s bottleneck [16], these solutions can become unsuitable for resource-constrained environments. Some FER researchers have developed lightweight approaches that aim to maintain a comparable level of performance while greatly reducing parameters and computational cost [1; 13; 26; 29; 40]. These proposals generally employ well-studied, lightweight CNN architectures and combine them with custom modules or training strategies to maintain robustness for FER ITW. Such methods have achieved great results, with some even setting the current state-of-the-art [29]. However, this is still only a small subset of FER research, and only an even smaller subset has been tested on ITW databases [40].

One such lightweight FER network is EfficientFace [40]. This network achieves great accuracy results on the RAF-DB (88.36%) and AffectNet (63.7%) datasets when taking into account its very low parameter count of 1.28 million.

However, other Transformer models like APViT [36] are capable of achieving much better results (RAF-DB 91.98%, AffectNet 66.91%), albeit with a significant increase in computational cost (65.2 M parameters).

This thesis proposes to bridge that accuracy gap by modifying the training procedure of EfficientFace rather than its deployed architecture. Specifically, we design a modified label distribution generator (M-LDG) that incorporates FER-pretrained backbone weights, local-global feature extraction modules, and knowledge distillation from APViT. The M-LDG acts as an enhanced teacher during training, generating richer soft targets for EfficientFace, while the deployed network remains unchanged. This approach attempts to transfer APViT’s representational capacity to EfficientFace without adding to its inference cost.

1.2. Document Structure

This document is organized as follows. The Theoretical Framework chapter surveys the deep learning foundations and FER background required to understand the proposed approach, covering convolutional networks, vision transformers, label distribution learning, and the key related architectures. The Hypothesis and Objectives chapter formalizes the research hypothesis, primary and specific objectives, and the scope and limitations of the study.

The Methodology chapter describes the two-phase factorial experimental design,

including factors, levels, response variables, hardware setup, and the statistical analysis procedures used to evaluate results. The Design chapter details the proposed algorithm: the architecture of the M-LDG, the two-stage training pipeline, dataset preprocessing, and implementation details.

The Results chapter reports the quantitative outcomes of both experimental phases, including ablation study findings, hyperparameter optimization results, and per-class performance metrics. The Discussion chapter interprets those results in light of the hypothesis, and identifies the capacity bottleneck as the primary failure mode.

Finally, the Conclusions and Future Work chapter draws the principal conclusions from this research effort and outlines directions for future work.

Theoretical Framework

“Read not to contradict and confute;
nor to believe and take for granted;
nor to find talk and discourse; but to
weigh and consider.”

Sir Francis Bacon

This chapter lays the theoretical groundwork required to understand the research presented in this thesis, building on the motivation established in Introduction. We survey the deep learning techniques and facial expression recognition approaches that inform our proposal, covering convolutional neural networks, attention mechanisms, vision transformers, label distribution learning, and the specific architectures, EfficientFace and APViT, that form the core of our experimental design. The chapter closes with a review of related lightweight FER work that contextualizes the contribution.

2.1. Deep learning in computer vision

Deep learning (DL) has revolutionized the field of computer vision (CV) by allowing automated feature extraction from images with little manual engineering required [17]. Following the description by LeCun et al. [17], DL is a subset of machine learning techniques that use neural networks (NNs) with several hidden layers to model data patterns. These multi-layered networks automatically learn representations of input data in increasing levels of abstraction. NNs are trained by processing the outputs for a dataset, calculating the error, and using its gradient to adjust the network’s weights or parameters in a direction that will minimize the error. Instead of one big adjustment, many smaller ones are performed iteratively as part of the stochastic gradient descent algorithm.

2.1.1. Convolutional neural networks (CNNs)

Convolutional neural networks (CNNs) are a frequently used method for computer vision tasks. CNNs earn their name from the convolution operation that happens within their layers. They are designed to learn representations from multi-dimensional input arrays. A relevant example is a color image which can be passed to the network

as three 2D arrays, one for each RGB channel [17].

CNN architectures repeatedly stack layers to add more depth to the network, with depth being an important factor in performance gain [12]. These layers serve as a powerful and automatic feature extraction component. For classification tasks, a classifier network is appended to learn how to categorize the high level representations. These classifiers usually take the form of fully-connected layer stacks [17].

CNNs can recognize patterns in images in a hierarchical manner, starting with edges and moving on to complex structures like objects. For classification tasks, these higher abstractions amplify more important aspects for discrimination [17]. Figure 2.1 shows the hierarchy of the features extracted by a CNN from a given RGB input image. Each feature map aligned horizontally corresponds to one of many output channels computed from the same input.

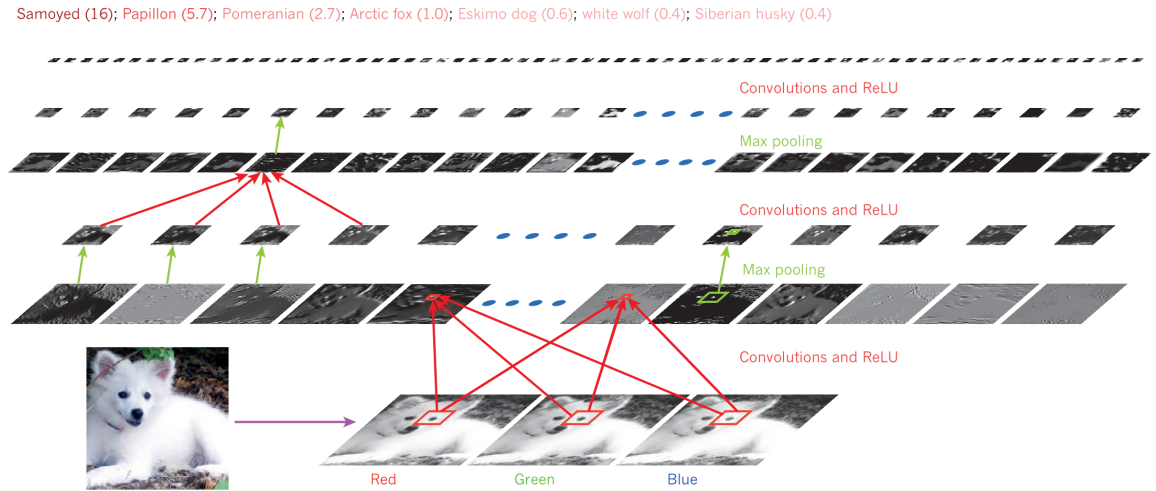
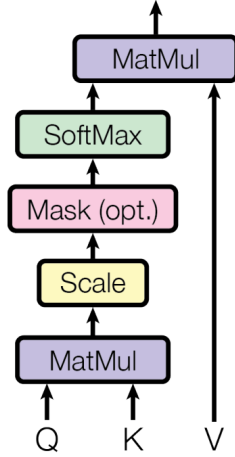


Figure 2.1: Hierarchy of CNN feature maps. Each row denotes the feature maps extracted by the same CNN layer. Note how each map or channel shows different types of features like edges and textures. Image source [17].

2.1.2. Self-attention

Scaled Dot-Product Attention



Multi-Head Attention

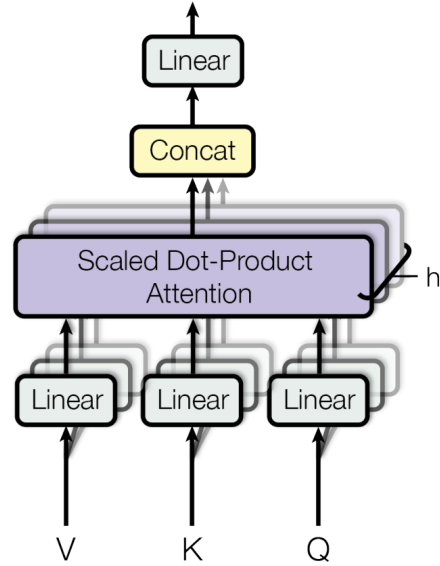


Figure 2.2: Diagram of scaled dot-product attention and multi-head self-attention operations. Image source [33].

In 2017, Vaswani et al. [33] published a seminal paper proposing a particular type of self-attention mechanism for sequence transduction neural networks. Self-attention mechanisms seek to find relations between different positions of a sequence to find a representation for it. Driven by existing applications of self-attention for natural language processing (NLP) tasks, [33] proposed **scaled dot-product attention**. From this point forward, we'll refer to this concept as simply self-attention for brevity.

Self-attention [33] is a function that maps a query (Q), and a key-value pair (K, V) to an output, with all of these components being vectors.

This is accomplished by computing a weighted sum of the values, where the weights are given by a compatibility function between queries and keys.

To allow self-attention to learn information at different representation levels, the authors propose multi-head self-attention (MHSA).

MHSA linearly projects the query, key, and value vectors h times, one for each attention head, and then computes self-attention for all the heads in parallel.

Figure 2.2 shows the standard self-attention and multi-head self-attention operations.

2.1.2.1. Transformers

Vaswani et al. [33] also presented an application of MHSA in the form of the Transformer architecture shown in figure 2.3. The Transformer is a network for the sequence transduction task that follows an encoder-decoder structure. Both the encoder and decoder employ stacked layers that each apply MHSA to a given input sequence. The encoder learns intricate representations of the input sequence, while the decoder takes these representations to generate a corresponding output sequence.

2.1.3. Deep computer vision architectures

We now mention three deep learning architectures in computer vision tasks. We include relevant networks that are part of later works focused on FER with deep learning.

2.1.3.1. Residual networks (ResNets)

He et al. [12] presented residual networks (ResNets) as a solution to the accuracy degradation problem that occurs when training very deep networks. ResNets employ a residual learning approach where the usual stack of CNN layers is enhanced by adding shortcut or skip connections. These connections add the unmodified output of a given layer to the output of another one located a few steps ahead. Figure 2.4 shows the basic residual learning block that forms a ResNet architecture.

The authors experiment with various network depths for the ImageNet [3]

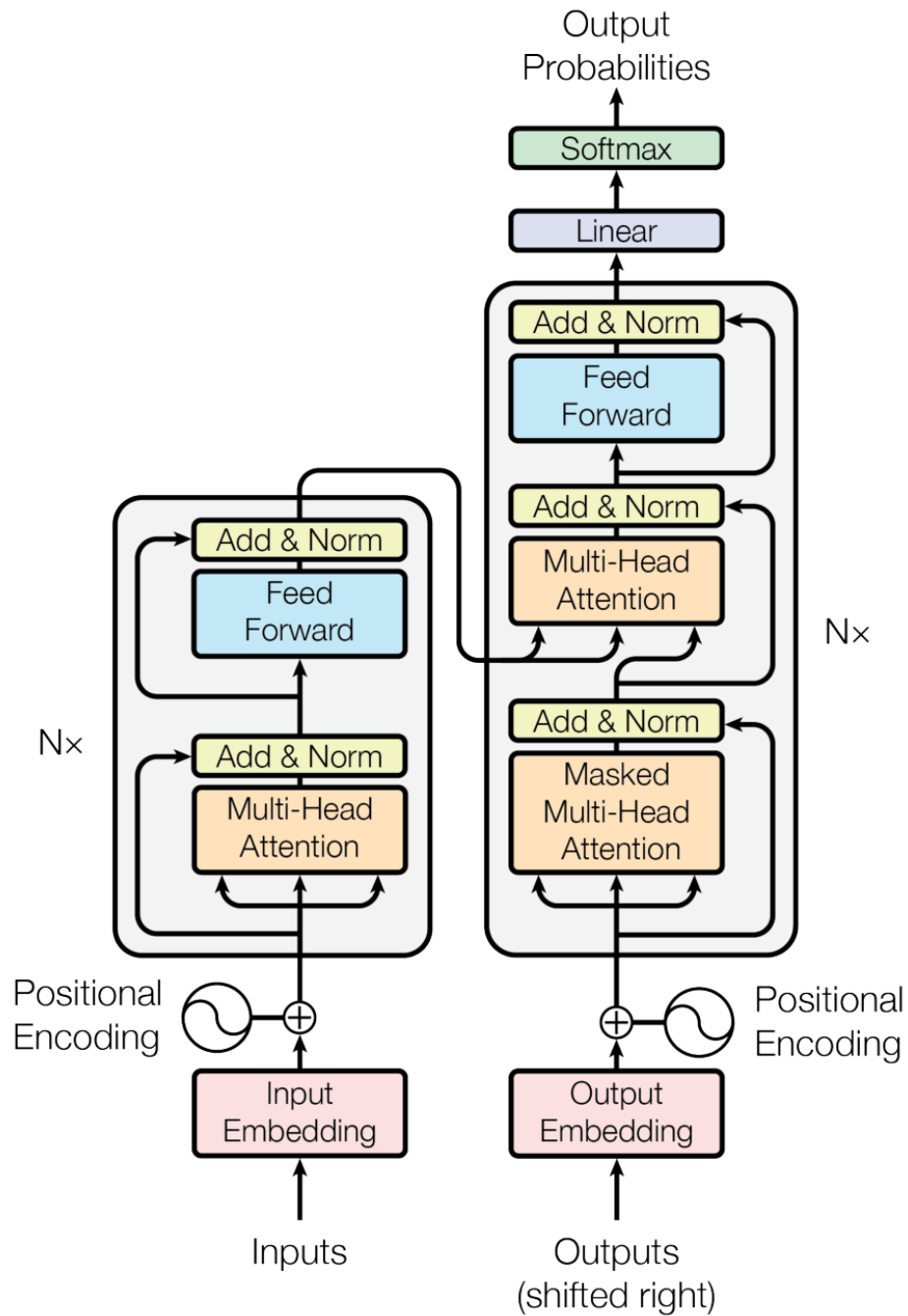


Figure 2.3: Transformer architecture. The encoder (left column) feeds its outputs to the decoder (right column) to generate output probabilities for the given sequence. Image source [33].

classification problem. By convention, a residual network is referred to as ‘ResNet’ followed by the number of layers, e.g. ResNet-50.

2.1.3.2. ShuffleNet

While very deep networks achieve impressive accuracy on classification and other tasks [12], it is unfeasible to employ them in mobile devices or other resource-constrained environments. In response to this, certain authors have focused their work on creating lightweight CNN models that reduce the number of parameters and floating point operations per second (FLOPs) while maintaining performance.

One such example is the ShuffleNet family of models, first introduced by Zhang et al. [38] in 2018. ShuffleNet uses a specialization of the convolution operation, known as pointwise group convolution, as well as a channel shuffling method to reduce the computational cost and memory usage.

Ma et al. [24] later proposed an optimized version of ShuffleNet known as ShuffleNet V2. The authors of ShuffleNet V2 depart from the intuition that FLOPs are an indirect metric for comparing efficiency, whereas real world speed is a much more accurate point of comparison. Under this assumption, they first devised a set of 4 guidelines for efficient network design. These guidelines are based on a careful experimental design using

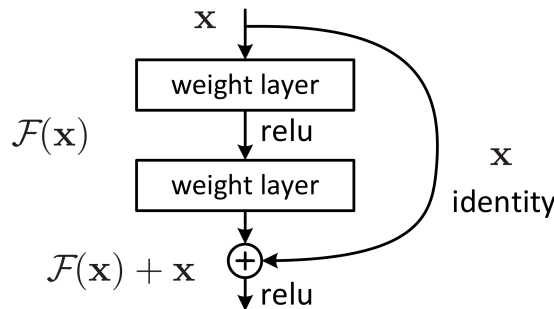


Figure 2.4: Residual learning block. Image source [12].

existing lightweight networks and real-world hardware platforms. Next, the authors analyzed which guidelines ShuffleNet was breaching, and proposed an optimized version that adhered to them. This resulted in a network with a much higher image throughput and slightly better accuracy than its predecessor.

2.1.3.3. Vision Transformer (ViT)

The Vision Transformer (ViT) by Dosovitskiy et al. [6] applies the Transformer [33] architecture to the ImageNet [3] classification problem. The authors employ only the encoder portion of the Transformer to generate embeddings for the input image. The architecture of the encoder block used in ViT is shown in the rightmost side of figure 2.5.

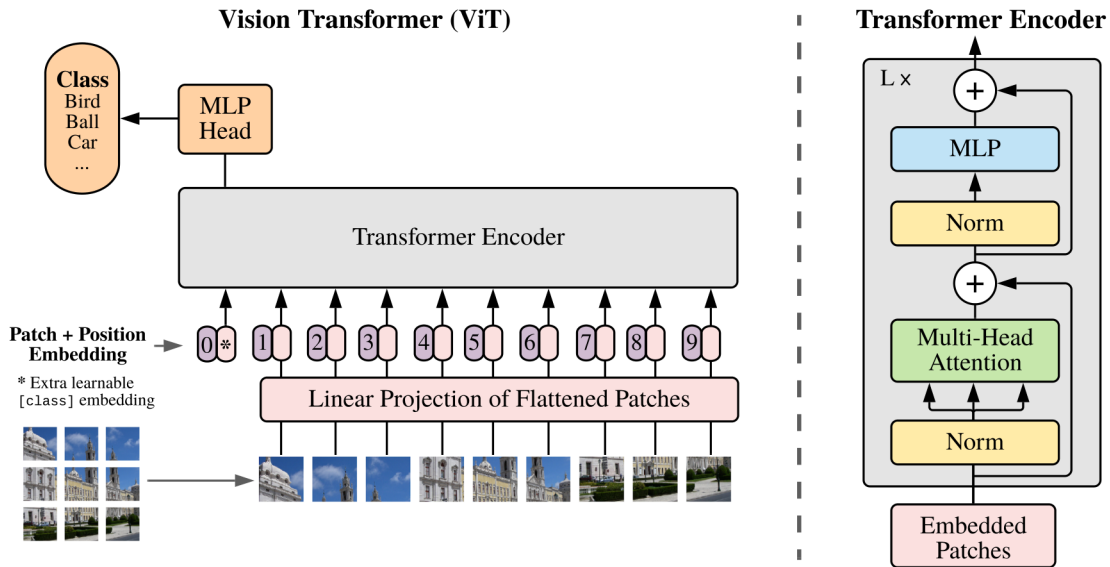


Figure 2.5: Vision Transformer (ViT) and Transformer encoder architecture. MLP denotes a fully-connected network. Image source [6].

In order to feed 2D images to the encoder, the authors split the images into patches of size 16×16 that are treated as a sequence of visual tokens. The patches are flattened into a vector, and linearly projected by a trainable network to obtain a

sequence of patch embeddings. Next, positional embeddings are added to the patch embeddings to retain positional information. The resulting sequence is prepended by a class token that, after being processed by the encoder, will contain a representation of the entire image. This special token is fed to a fully-connected classifier head to predict output probabilities. Figure 2.5 shows the ViT architecture, as well as the previously mentioned processing of inputs.

A caveat of ViTs is that they may achieve similar or inferior performance when compared to established models like ResNets. However, when pretrained with sufficiently large datasets, they can outperform such methods. Another advantage is that the self-attention operation leads the model to pay attention to image regions that are more relevant for classification, e.g. tune out the background [6].

In the remaining pages of this chapter, we will discuss the facial expression recognition problem with a focus on the deep learning methods we previously discussed.

2.2. Facial expression recognition (FER)

Facial expression recognition (FER) is a field within affective computing that seeks to detect human emotions based on expressions [16]. Facial expression is a non-verbal form of communication that can relay a wide variety of psychological information in a rapid manner [31]. Additionally, facial expression information can be captured in a non-invasive way by visual methods [32]. As such, FER has many applications within medicine, driver safety, and human-computer interaction [20].

Early FER researchers applied classical image processing approaches such as

hand-crafted features coupled with support vector machines or dynamic Bayesian networks for suitable classification performance [2]. However, with the advancements in neural network research, processing power, and data gathering capabilities, deep learning-based approaches have become the norm in FER [2; 16; 20].

Recent secondary studies have identified the general deep FER system pipeline to employ three main steps: (i) facial image preprocessing, (ii) feature extraction, and (iii) classification [16; 20]. Preprocessing usually involves the following methods: illumination normalization, data augmentation, face detection and alignment, and frame normalization for video-based FER [16]. Currently, researchers have developed deep learning approaches for all three stages. However, step (i) is usually applied independently by using methods such as [37] and [5] for face detection and alignment prior to training the feature extractor and classifier end to end. Such an approach can be seen in recent works such as [23; 35; 36; 40]. In this research proposal, we shall focus on the latter two steps exclusively.

The following sections characterize our particular area of research within FER.

2.2.1. Theories for emotional labeling

As part of a survey on deep learning techniques for FER, Karnati et al. [16] identified two main theories for representing emotional states: dimensional and discrete. In the dimensional theory, an emotion is placed in the continuous, 2D valence-arousal space. Figure 2.6 shows the labeling employed under this model.

The discrete theory is based on Ekman and Friesen’s [7] universal emotions model, which defines six basic categories of expression: anger, disgust, fear, happiness, sadness, and surprise. While it is argued that this model is prevalent across cultures, later

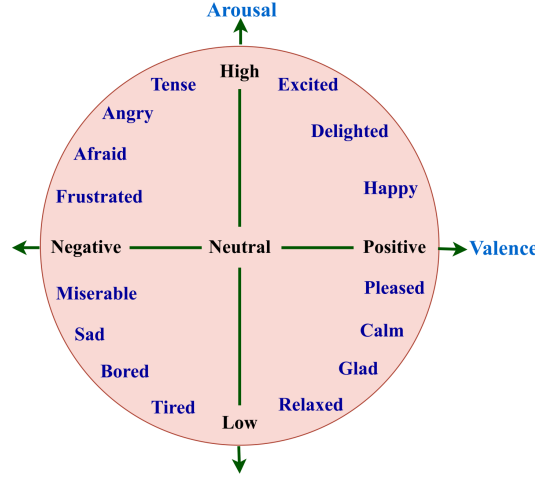


Figure 2.6: Dimensional model for emotional labeling. Image source [16].

research seems to suggest the contrary [20]. Regardless, the discrete theory remains the dominant model for FER research [16]. The discrete theory is also often extended by adding the neutral and contempt categories such as in [27]. This research proposal focuses on the discrete theory for emotional labeling.

2.2.2. Static FER

Authors such as Li et al. [20] and Karnati et al. [16] categorize FER approaches by the type of input given to the feature extractor. Our proposal is intended to be a part of static FER research, which accepts single facial images as input and extracts features from spatial information exclusively. On the other hand, dynamic FER adds a temporal component by accepting image sequences or videos as input.

A large amount of research has focused on static FER due to availability of data and ease of processing [20]. Additionally, dynamic FER databases can be used for static research by simply extracting single frames from the sample videos.



Figure 2.7: Comparison of lab-controlled (CK+ [22]) and ITW (AffectNet [27]) database samples for the six basic emotion classes.

2.2.3. FER in-the-wild (ITW)

Another important change in FER research over the years is the shift from laboratory-controlled conditions to in-the-wild (ITW) datasets. This shift has occurred in parallel with the transition from classical machine learning methods to deep learning approaches [16; 20]. While DL methods achieve excellent accuracy in lab-controlled data, their performance for FER ITW is far inferior [23; 39]. Despite this, FER systems trained in lab conditions are less suitable for real-world use thanks to a lessened generalization capability [21]. This proposal focuses on FER ITW as part of its research problem.

In ITW databases, factors such as motion blur, face deformation, head pose, facial occlusion, and illumination can vary significantly. Such uncontrolled conditions can lead to a vastly degraded performance of FER systems [23]. Figure 2.7 shows a comparison between lab-controlled and ITW samples.

Lab datasets typically contain significantly fewer images when compared to their ITW counterparts [23]. For instance, CK+ is a lab-controlled dataset frequently used for benchmarking FER systems [16]. CK+ [22] comprises 593 facial expression

sequence samples, with only 327 labeled with a discrete emotion class. In contrast, the ITW dataset, AffectNet [27], includes approximately 1 million facial images, with 450 thousand manually labeled with a discrete class.

2.2.4. Challenges in FER

The transition to ITW datasets for FER has been beneficial for robust training, but it has also introduced several challenges in system design. According to a recent survey [16], pose and occlusion are two primary barriers to the adaptability of such systems. Said survey [16] also mentions annotator subjectivity, or label noise, as a factor that leads to incorrectly labeled images. Additionally, FER ITW datasets exhibit high variation within classes due to differences in age, gender, and ethnicity, as well as high similarity across classes, such as fear and surprise [8; 9; 20].

2.2.5. Label distribution learning (LDL)

Label distribution learning (LDL) was originally introduced by Xin Geng [10] as a general learning paradigm for machine learning. LDL was later adapted to the FER problem by Zhou et al. [42] in the form of emotion distribution learning (EDL).

The authors of EDL depart from the intuition that facial expressions are often the result of a mixture of basic emotions, instead of a particular one. EDL is proposed as an alternative to single-label learning (SLL) and multi-label learning (MLL) that can capture the intensity of each emotion in a facial expression. The intensity is given by a description degree d_x^y , where x is a sample in the dataset and y is a particular emotion class. The degrees associated with a sample x are arranged in an emotion distribution $E_x = \{d_x^{y_1}, d_x^{y_2}, \dots, d_x^{y_c}\}$, where c is the number of classes. The degrees are normalized such that $d_x^y \in [0, 1]$, and the sum of all degrees for a sample is equal to one. Under EDL, the error of a FER model can be computed by any function that

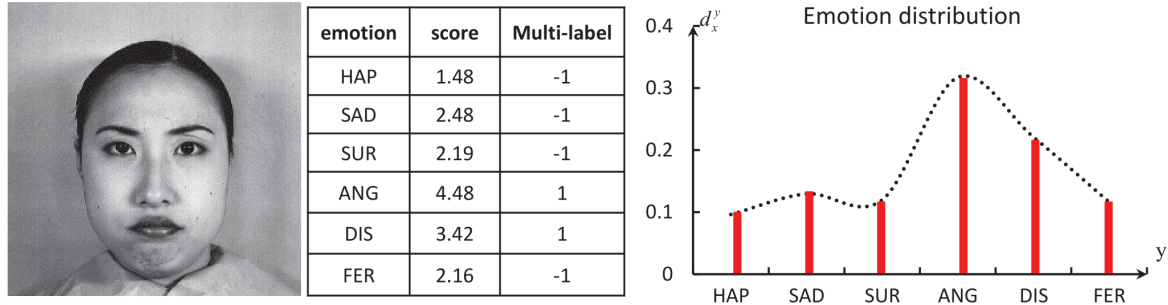


Figure 2.8: Emotion distribution of a facial image as well as class scores and multi-labels. Image source [42].

measures the distance between two distributions.

Figure 2.8 shows the three types of labels for a facial expression image. Here, a SLL approach will classify the expression as anger based on the highest score and a MLL approach will detect anger and disgust. Similarly to MLL, EDL indicates the presence of anger and disgust. However, the description degrees also tell us that anger has a higher intensity than disgust and is potentially more relevant.

2.3. Deep FER architectures

The following sections describe two architectures for static FER ITW that are central to our proposal.

2.3.1. EfficientFace

As a solution to resource-intensive architectures, Zhao et al. [40] proposed the EfficientFace (EF) architecture for lightweight FER. In order to adapt the ShuffleNet V2 [24] architecture towards FER, the authors adopt two main perspectives: improved feature extraction, and label distribution learning. Figure 2.9 shows the architecture of the EF network.

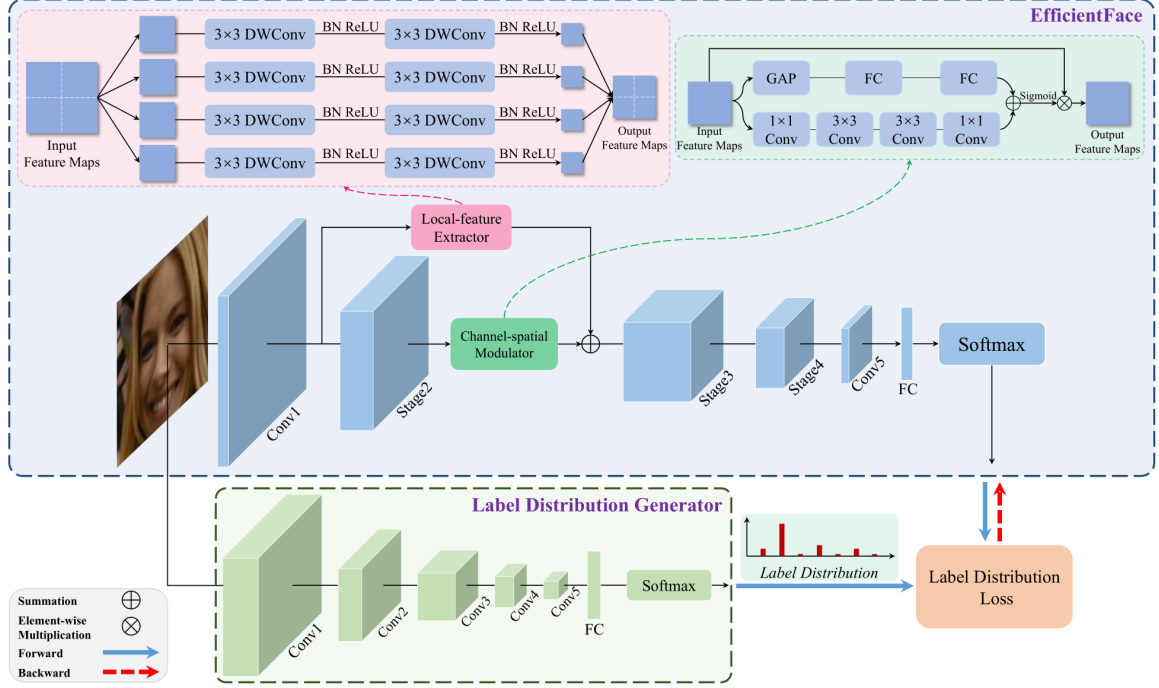


Figure 2.9: EfficientFace and label distribution generator (LDG) architecture. DWConv denotes depthwise convolution, GAP denotes global average pooling, and FC denotes a fully-connected layer. Image source [40].

The improved feature extraction comes in the form of two specialized modules that are adapted into ShuffleNet V2 [24]. The first module is the local-feature extractor (LFE), which learns local features at the low feature level. The LFE accepts the output features from the first convolutional layer of ShuffleNet V2 and splits them spatially into four equal patches. These patches correspond to the facial units related to expression. Each patch is processed in parallel by two stacks of convolutional layers and then concatenated to obtain a set of **local feature maps**. In order to maintain computational efficiency, the LFE employs depthwise convolutions.

The next specialized module is the channel-spatial modulator (CSM). The CSM is introduced after stage 2 to essentially create an attention map of features at a global level. To accomplish this, the CSM employs two parallel branches to compute two sets of heatmaps from spatial and channel information. The computed channel and spatial

heatmaps are fused together, and then multiplied with the CSM input features to apply the attention maps. This last step yields what is known as the **modulated global feature maps**.

The next step is to take the local and global feature maps and add them with a residual connection as seen in figure 2.9. The remaining portion of EF takes the resulting local-global features and processes them using the rest of the ShuffleNet V2 layers without further modification. The final output layer is a softmax function which computes an emotion distribution for the given sample.

The authors conducted an ablation study to test the effectiveness of the proposed modules. On both evaluation datasets, their results show that each module improves the network’s accuracy individually. Furthermore, the combination of both modules (as seen in figure 2.9) provides the highest increase in accuracy.

2.3.1.1. Label distribution generator (LDG)

The label distribution generator (LDG) is the component used to implement the second perspective: label distribution learning (LDL). The LDG is a standard ResNet-50 [12] network pretrained on a large dataset of celebrity facial images [11], and fine-tuned for the FER task. The network is trained using the typical single-label learning method (SLL), with a final softmax function for computing the output emotion distribution. The LDG architecture is presented in the bottom part of figure 2.9

Once the LDG is trained, it is set to inference mode and used to guide the training of EfficientFace (EF) using LDL. During training, both networks generate their respective emotion distributions. Next, a cross-entropy function is used to measure

the distance between distributions, using the LDG’s output as ground-truth.

Once training is finished, the LDG can be discarded to yield a very light network with competitive FER performance. One important result of the authors’ study is that EF trained with LDL obtains a better accuracy than the standalone LDG, despite being a much smaller network.

2.3.2. APViT

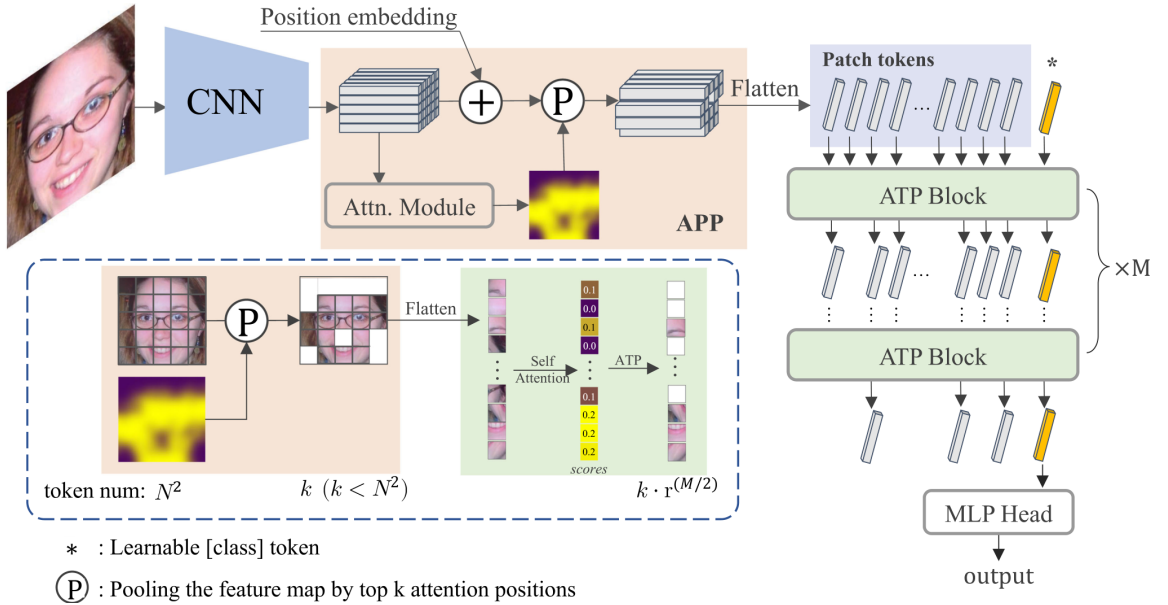


Figure 2.10: APViT architecture. The dashed rectangle presents an alternative view for the processing of inputs. APP denotes attentive patch pooling, ATP denotes attentive token pooling, MLP denotes a fully-connected network, M denotes the number of stacked ATP blocks, and r denotes a configurable keep rate for tokens. Image source [36].

The self-attention mechanism in Vision Transformers (ViTs) can reduce the impact of noise like pose and occlusion variations. However, many unimportant features are still processed. To address this, Xue et al. [36] proposed the APViT model for FER ITW. They introduced two attentive pooling modules to discard irrelevant information and reduce FLOPs. Figure 2.10 shows the architecture of the network.

APViT uses a ResNet-50 variant, IR-50 [4], as a backbone to extract low-level feature maps from an input image. The maps are split into patches, and infused with positional embeddings. Each patch’s attention score is calculated and arranged in a grid. The first module, attentive patch pooling (APP), selects the top-k relevant patches based on the attention grid, discarding the rest. These patches are flattened into tokens and fed to the ViT portion of the network. The second module, attentive token pooling (ATP), is appended to each Transformer encoder layer, forming ATP blocks. ATP retains only the top-k important tokens using self-attention scores, thus reducing the number of tokens the encoder outputs.

APViT’s pooling modules reduce FLOPs by 17% compared to the baseline. They also improve performance over its predecessor, TransFER [35], with accuracy gains of 1.07% and 0.68% on the RAF-DB [19] and AffectNet [27] datasets, respectively. However, APViT retains the same 65.2 million parameter count as TransFER, making it a resource-intensive model [29].

2.4. Related Work

The following section describes various deep learning-based approaches for static FER. Given the nature of our proposal, we focus on lightweight models and those that implement self-attention mechanisms such as vision transformers.

2.4.1. Lightweight FER models

Earlier lightweight FER studies tended to adapt existing CNN modules or architectures towards the FER problem. Hewitt and Gunes [13] performed a comparative study between FER architectures for mobile environments focusing on

small model sizes. The study proposed and compared three network variants based on popular CNN architectures: AlexNet, VGGNet, and MobileNet. Mitre-Hernández and Ferro-Perez [26] proposed ResMoNet, which stacks different optimized CNN blocks for lightweight FER. Their approach achieves a smaller size and lower FLOPs than MobileNet, with a lower memory utilization than other comparable networks. FaceChannel, proposed by Barros et al. [1], is a FER model that predicts emotions based on the dimensional, valence-arousal model instead of the categorical one. It is a small network based on VGG16 that trims down the number of layers to obtain a significantly smaller number of parameters.

More recent studies have combined the current versions of lightweight models such as ShuffleNet and MobileNet with custom modules to achieve outstanding performance. EfficientFace by Zhao et al. [40] integrates specialized modules for local-global feature extraction into the lightweight ShuffleNet-V2 architecture. Additionally, the authors adopt a label distribution learning approach where they train EfficientFace to learn the outputs of a FER-trained ResNet-50 instead of the usual dataset labels. Wu et al. [34] also propose a lightweight FER network based on ShuffleNet-V2 intended for a cloud-edge device model. Their approach proposes a system that runs the lighter network components on edge devices. However, as an optional step, the intermediate feature maps are sent to a more powerful cloud device that computes channel-spatial attention maps with custom pooling modules. PAtt-Lite by Le Ngwe et al. [29] instead uses a truncated MobileNetV1 as its backbone. This truncated network is followed by a patch extraction block that splits the feature maps into four patches for local feature extraction. Finally, the patch feature maps are fed to a special module that uses self-attention for classification.

2.4.2. Self-attention in FER

To the best of our knowledge, Xue et al. [35] were the first group of authors to apply Vision Transformers (ViTs) to the FER problem. They propose the TransFER model, which applies a specialized type of dropout at different stages to help the model learn more diverse representations. The attention dropping modules are applied at the convolutional and ViT encoder stages to randomly drop feature maps or self-attention heads respectively. Huang et al. [15] presented a network that uses an attention mechanism at both the low and high levels of feature learning. At the low level, a grid-wise attention mechanism is used to learn long-range biases between different facial features. At the high level, the feature maps are transformed into visual tokens and fed to a ViT encoder that learns global representations of expressions. Ma et al. [23] instead propose feeding richer feature maps to the ViT encoder network by diversifying the network inputs. Specifically, they extract the local binary pattern (LBP) from the input image and extract features from both the RGB and LBP data using two parallel ResNet-18 networks. These separate maps are fused together by a special module that also extracts local and global features and then feeds them to the ViT encoder. Iterating on the TransFER model [36], Xue et al. propose the APViT network. They employ two attention pooling mechanisms to drop feature map patches and visual tokens based on their attention scores, discarding the less important regions. In doing this, they manage to significantly lower the model’s FLOPs while improving accuracy by avoiding noisy features.

A smaller number of authors have also applied the self-attention mechanism for FER without using a ViT encoder component. Pecoraro et al. [30] propose a special module that applies multi-headed self-attention to the output feature maps of convolutional layers. This module is intended to be compatible with existing networks so that it only has to be inserted between the CNN layers without requiring further modification. The

attention classifier in PAtt-Lite [29] also employs dot product self-attention on the last fully-connected layers of the network. The ablation study shows that this attention classifier obtains significant performance improvements when applied.

Hypothesis and Objectives

“The first principle is that you must not fool yourself — and you are the easiest person to fool.”

Richard P. Feynman

3.1. Hypothesis

Based on the literature review and objectives of this research project, we present the following hypothesis:

“Training EfficientFace with a label distribution generator enhanced with one or more of: FER pretrained backbone weights, local-global feature extraction modules, and/or vision transformer guidance, should increase its facial expression recognition accuracy on the AffectNet dataset when compared to the original baseline”.

3.2. Objectives

This section defines the main and specific objectives of our research proposal. These objectives align with the complete execution of our proposed study.

3.2.1. Main Objective

Based on our proposal to address the previously presented problem, we determine the main objective of our research project to be as follows:

Study the facial expression recognition accuracy obtained when training EfficientFace [40] with a label distribution generator enhanced with local-global feature extraction modules and knowledge from a vision transformer model [36].

3.2.2. Specific Objectives

In order to successfully complete our general objective, we derive the following specific objectives:

1. Enhance the label distribution generator (LDG) by loading the pretrained weights from APViT’s backbone, adding local-global feature extraction modules

from EfficientFace, and training the LDG to imitate APViT’s outputs using label distribution learning.

2. Study the performance changes induced by each modification to the LDG by means of ablation study and determine the best combination.
3. Train two EfficientFace variants, one using the modified LDG and one using APViT, and study the performance differences with respect to the baseline and other state-of-the-art approaches.

3.2.3. Deliverables

This section defines the deliverables aligned with each of the specific objectives of the thesis study.

Specific Objective 1:

- Python script that defines the modified LDG network architecture and its training procedure.
- Preprocessed AffectNet dataset.

Specific Objective 2:

- Jupyter notebooks that automate the Phase I experiment execution.
- Raw data results from the Phase I experimental design execution.
- Raw data results from the Phase I hyperparameter optimization procedure.
- Trained weights of the resulting models.

Specific Objective 3:

- Jupyter notebooks that automate the Phase II experiment execution.
- Raw data results from the Phase II experimental design execution.

- Raw data results from the Phase II hyperparameter optimization procedure.
- Trained weights of the resulting models.

3.3. Scope and Limitations

The following section outlines the scope and limitations of this research project. These are the specific boundaries set for the study, which limit the scope and define the operational framework within which the research was conducted.

- This research focuses exclusively on the static facial expression recognition task with images annotated under the 7 or 8 class discrete model.
- This project does not extend to adjacent research areas such as facial detection, facial identification, dynamic (video) FER, micro-expression recognition, or FER with facial action units.
- The preprocessing steps for the input images was carried out as outlined by the authors of EfficientFace and APViT in their respective papers [36; 40].
- The materials used in this research are limited to publicly available resources. This includes datasets, research papers, source code, programming languages, packages, and other language extensions.
- Only deep learning methods for facial expression recognition were considered for the state-of-the-art comparison.

Methodology

“If you cannot measure it, you
cannot improve it.”

Lord Kelvin

In this chapter we describe the key aspects of the methodology used to test our hypothesis and proposal.

We go over the two phase design of the study, along with their factors, levels, responses, inputs, statistical analysis and more.

4.1. Experimental design

The experimental design of this study is divided into two distinct phases to ensure a rigorous and replicable process. This division allows for a more thorough evaluation of the proposed modifications and a reduction in overall training time. The design for each of the phases follows the guidelines defined by Montgomery [28].

4.1.1. Phase I: LDG ablation study

The first phase focuses on analyzing the performance changes induced by modifying the label distribution generator (LDG) network that guides EfficientFace’s training. An ablation study is conducted to evaluate the effects of each proposed modification to the LDG, concentrating solely on the accuracy response for the facial expression recognition (FER) task. EfficientFace is not trained or evaluated in this phase. The best modified LDG configuration identified here will be used in phase II.

4.1.2. Phase II: EfficientFace training study

The second phase analyzes the effectiveness of using the modified LDG as an intermediary to transfer knowledge from APViT to EfficientFace. The best LDG configuration from phase I is used to train EfficientFace, following the original procedure [40]. Additionally, an alternative EfficientFace variant is trained using APViT directly as its label distribution generator, skipping the modified LDG. A comparative experiment is conducted to determine the best approach.

4.2. Factors and levels

The factors and levels for each phase of the experimental design are summarized below.

4.2.1. Phase I: LDG Ablation study

The design factors for this phase are the three proposed modifications to the LDG (pretrained weights, local-global feature extraction, and transformer guidance), as well as the training dataset. Since this is an ablation study, the levels of the factors are binary (present or not present). The factors and levels are summarized in Table 4.1.

	Factors			
	Pretrained Weights	GL Feature Extraction	Transformer Guidance	Dataset
Levels	Yes	Yes	Yes	AffectNet (7 class)
	No	No	No	AffectNet (8 class)

Table 4.1: Factors and levels of phase I

4.2.2. Phase II: EfficientFace training study

The design factors for this phase are the guidance network (modified LDG or APViT) and the dataset (AffectNet 7-class or 8-class subset). The factors and levels are summarized in Table 4.2.

	Factors	
	Guidance Net	Dataset
Levels	M-LDG	AffectNet (7 class)
	APViT	AffectNet (8 class)

Table 4.2: Factors and levels of phase II

4.3. Measures and combinations of the experiment

The measures and treatment combinations for each phase are described below.

4.3.1. Phase I: LDG ablation study

A 2^k factorial design with $k = 4$ is used to explore the effects of each LDG modification and their interactions, with both AffectNet subsets. This results in 8 treatment combinations per subset, for a total of 16 experimental runs. The treatment combinations are shown in Table 4.3.

For reference, the factors are pretrained weights (A), GL feature extraction modules (B), APViT guidance (C), and dataset (D).

Run	A	B	C	D	Label
0	+	+	+		abc
1	+	+	+	+	abcd
2	+	+			ab
3	+	+		+	abd
4	+		+		ac
5	+		+	+	acd
6	+				a
7	+			+	ad
8		+	+		bc
9		+	+	+	bcd
10		+			b
11		+		+	bd
12			+		c
13			+	+	cd
14					(1)
15				+	d

Table 4.3: Treatment combinations for phase I.

4.3.2. Phase II: EfficientFace training study

A 2^k factorial design with $k = 2$ is used, resulting in 4 treatment combinations executed in randomized order. Additional runs may be added based on hyperparameter

variation. The treatment combinations are shown in Table 4.4.

The factors in this phase are labeled: guidance network (G), and dataset (D).

Run	G	D
1	LDG	AffectNet (7 class)
2	LDG	AffectNet (8 class)
3	APViT	AffectNet (7 class)
4	APViT	AffectNet (8 class)

Table 4.4: Treatment combinations for phase II

4.4. Response variable

For both phases of this study, the top-1 accuracy metric obtained on the evaluation or test set after executing the full training procedure is taken as the sole response variable. This aligns with common practice in FER studies, where accuracy is the main metric for comparison with other approaches [2; 16; 20].

Top-1 accuracy measures the fraction of samples where the model’s most confident prediction exactly matches the true label:

$$\text{Top-1 Accuracy} = \frac{1}{N} \sum_{i=1}^N [y_i = \hat{y}_i] \quad (4.1)$$

where N is the total number of samples, y_i is the true label, and $\hat{y}_i$ is the predicted label with the highest probability.

4.5. Experiment set up and data collection

The execution of the experiments in both phases is facilitated by Jupyter notebooks that automate set up and parameterization.

For each phase there are $n = 2^k$ notebooks, each with the correct treatment combinations according to tables 4.3 and 4.4.

The notebook uses the levels to download the correct dataset, pretrained weights, and training script.

The script is then executed using the levels corresponding to the actual combination.

The randomization of runs and execution of the actual notebook is manually performed in the execution environment.

Multiple VastAI instances with a single Nvidia RTX 3090 GPU were used as the execution environment, isolating each run in its own instance.

The data collection is also automated via Wandb integration in the training scripts.

All experimental runs are logged to this service, registering accuracy, loss, confusion matrices, and hyperparameters, among other data.

4.6. Description of the inputs of the system

The main inputs to the system are the AffectNet dataset and the network weights for APViT and EfficientFace. All images are preprocessed using RetinaFace [5] for face alignment and maintaining the pipeline used in the original scripts of APViT [36] and EfficientFace [40].

4.7. Statistical tool

To analyze the results of both experimental phases, we use analysis of variance (ANOVA) to determine if there are statistically significant differences between the treatment combinations. The procedure is as follows:

First, we check the assumptions required for ANOVA, such as normality of residuals and homogeneity of variance, using diagnostic plots and statistical tests. If the assumptions are not met, appropriate data transformations or non-parametric

alternatives are considered.

For Phase I, after estimating the effects of each factor, negligible factors are discarded to focus the analysis on the most relevant variables. This also has the advantage of effectively adding replications to the remaining factor combinations, improving the reliability of the results. In Phase II, all factors are retained since multiple replications are available for each combination.

Descriptive statistics and visualizations, such as boxplots and line plots, are used to summarize the results and explore trends. Finally, a two-way ANOVA is performed to assess the main effects and interactions between the key factors, helping to identify which modifications lead to significant improvements in accuracy.

4.7.1. Hypothesis of the statistical tool

Null hypothesis: There is no statistically significant difference in the mean accuracy between the different treatment combinations; any observed differences are due to random variation.

Alternative hypothesis: At least one treatment combination leads to a statistically significant difference in mean accuracy compared to the others, indicating that the modifications or factors have a real effect on performance.

Design

“The details are not the details.
They make the design.”

Charles Eames

This chapter details the design of the algorithm used to test the hypothesis of this thesis. We begin by describing the architectures employed in our experiments, followed by a thorough explanation of the modifications and algorithmic strategies that underpin our proposal.

5.1. Description of architectures used to test our hypothesis

This section describes the two main deep learning architectures that form the foundation of our proposal: APViT and EfficientFace. These networks were selected due to their complementary strengths: APViT achieves state-of-the-art accuracy on facial expression recognition, while EfficientFace provides an efficient and lightweight architecture suitable for resource-constrained environments. For the sake of brevity, we describe the most relevant aspects of each network in regard to our proposal. For a more detailed description, refer to sections 2.3.1 and 2.3.2.

5.1.1. APViT: Attention Pooling Vision Transformer

APViT, proposed by Xue et al. [36], is a Vision Transformer architecture specifically designed for facial expression recognition. Building upon their earlier TransFER model [35], APViT introduces two key attention pooling mechanisms that significantly improve both accuracy and computational efficiency.

The network employs attention pooling at two stages: first, at the feature map level to discard less informative spatial patches, and second, at the visual token level to filter out noisy representations before the final classification stage. This dual pooling strategy allows APViT to focus computational resources on the most expressive facial regions while avoiding features that may introduce noise or

ambiguity.

APViT achieves state-of-the-art performance on major FER benchmarks, reporting 91.98% accuracy on RAF-DB and 66.91% on AffectNet [36]. However, this performance comes at the cost of a relatively large model size of approximately 65.2 million parameters, making it unsuitable for deployment in mobile or edge computing scenarios.

5.1.2. EfficientFace

EfficientFace, proposed by Zhao et al. [40], addresses the need for accurate yet lightweight facial expression recognition models. The architecture builds upon the efficient ShuffleNet V2 backbone and integrates specialized local-global feature extraction modules designed specifically for capturing both fine-grained facial details and holistic expression patterns.

The key innovation of EfficientFace lies not only in its architecture but also in its training methodology. Rather than training directly on dataset labels, EfficientFace employs a label distribution learning approach (LDL). In this framework, a label distribution generator (LDG), first generates soft probability distributions over expression classes for each training image. EfficientFace is then trained to approximate these distributions rather than the original hard labels. This emotion distribution learning process enables the lightweight network to learn more nuanced decision boundaries and achieve higher accuracy despite its compact size.

One of EfficientFace’s primary contributions is the balance between accuracy and efficiency, achieving competitive performance on FER benchmarks while maintaining a model size of only 1.28 million parameters [40]. This makes it particularly suitable for

real-world applications where computational resources are limited.

5.2. Dataset

We employ the AffectNet dataset by Mollahosseini et al. [27] as the dataset for the execution of our experimental design. Since EfficientFace [40] and APViT [36] both test on AffectNet, we opted to use this database as a fair point of comparison. Additionally, the dataset includes both 7 and 8 emotion class subsets. The 7 class subset can be taken as an easier problem than the 8 class one, with EfficientFace achieving an accuracy of 63.7% and 59.89% respectively. This results in a better distinction of the levels for this factor.

AffectNet is one of the largest FER datasets currently available, containing approximately 1 million images. This set was gathered automatically by querying search engines with emotion related keywords. The images are annotated under the categorical model with 11 emotion classes, as well as with valence-arousal values. Out of the total set of images, only 291,651 images are actually labeled with an emotion class. We have elected to follow the same procedure as [36; 40] for preprocessing, training, and evaluating of AffectNet. The classes for the 7 emotion subset correspond to neutral, happiness, sadness, surprise, fear, disgust, and anger, with the 8 class subset adding the contempt category.

5.3. Loss function for label distribution learning

In all training procedures we employ the same loss function as described in the original EfficientFace paper [40], which is the cross-entropy loss between the predicted distribution and the ground-truth distribution. This function is defined as:

$$\mathcal{L} = -\frac{1}{N \times c} \sum_{i=0}^{N-1} \sum_{j=0}^{c-1} d_j^i \log(\tilde{d}_j^i) \quad (5.1)$$

where N is the number of samples, c is the number of classes, $\tilde{\mathbf{d}} = (\tilde{d}_1, \tilde{d}_2, \dots, \tilde{d}_{c-1})$ is the predicted label distribution of EfficientFace and $\sum_{j=0}^{c-1} \tilde{d}_j = 1$; the superscript i and subscript j denote sample and expression category respectively.

5.4. Proposed algorithm

Our proposal aims to enhance EfficientFace’s performance by modifying its training procedure to leverage knowledge from the more powerful APViT model. Rather than directly using APViT as the label distribution generator, we propose creating a modified, intermediate LDG network that combines architectural components and knowledge from both APViT and EfficientFace.

5.4.1. Design of the proposed algorithm

The proposed algorithm consists of two main stages: (1) training a modified LDG that incorporates enhancements from both APViT and EfficientFace, and (2) using this modified LDG to train an improved EfficientFace variant through label distribution learning.

5.4.1.1. Modified Label Distribution Generator Architecture

The modified LDG is based on the same ResNet-50 architecture that serves as the feature extraction backbone of APViT. Three key modifications are proposed to enhance the LDG’s FER performance and in turn the performance of the resulting EfficientFace variant. The design for this stage is depicted in figure 5.1.

1. FER-pretrained backbone weights from APViT: Instead of initializing the ResNet-50 backbone with ImageNet [3] pretrained weights (as in the original

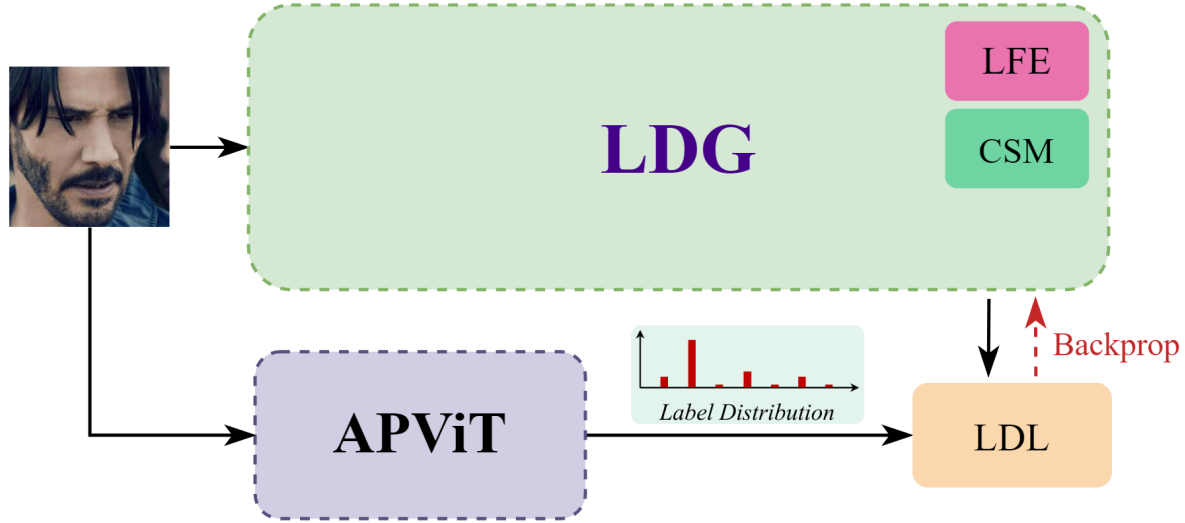


Figure 5.1: Training pipeline for the first phase of the proposed algorithm. The IR-50 [4] network is loaded with the backbone weights of APViT [36] and enhanced with the local-global feature extraction modules from EfficientFace [40]. It is then trained using APViT as the distribution generator.

EfficientFace approach), we initialize it with the backbone weights from a fully trained APViT model. Since APViT uses a hybrid architecture that includes convolutional layers before the transformer encoder, these pretrained weights are already specialized for facial expression recognition. This initialization provides the LDG with feature extractors that are tuned to capture expression-relevant patterns.

2. Local-Global feature extraction modules: We integrate the specialized local-global feature extraction modules from EfficientFace into the modified LDG architecture. These modules are inserted between the convolutional stages of the ResNet-50 backbone, allowing the network to extract both local facial features (such as subtle changes in specific facial regions) and global contextual information (such as overall facial configuration). This architectural enhancement helps the LDG generate more informative label distributions that capture the multi-scale nature of facial expressions.

3. Knowledge distillation from APViT: The modified LDG is trained using knowledge distillation, where the training objective is to minimize the difference

between the LDG’s output probability distributions and those generated by the fully trained APViT model. Specifically, we employ the same label distribution learning loss from EfficientFace to measure the discrepancy between the two distributions. This training procedure enables the modified LDG to learn to produce soft labels that approximate APViT’s predictions while maintaining a more efficient architecture.

5.4.1.2. Training EfficientFace with the Modified LDG

Once the modified LDG is trained, it serves as the label distribution generator for training EfficientFace. The training procedure follows the original label distribution learning approach [40], but with the modified LDG replacing the original ResNet-50 LDG.

During EfficientFace training, the modified LDG remains frozen and is used to generate soft label distributions for each training image. EfficientFace is then trained to minimize the loss between its predicted distributions and these soft labels. This two-stage knowledge transfer process, first from APViT to the modified LDG, then from the modified LDG to EfficientFace, should enable the lightweight model to benefit from APViT’s superior representational capacity without the computational burden of using APViT directly during training. The design for this stage is depicted in figure 5.2.

We employ a slightly tweaked version of the EfficientFace source code that fixes some minor bugs and limitations of the original implementation. The modified codebase is publicly available at <https://github.com/DavidCrgh/EfficientFace>.

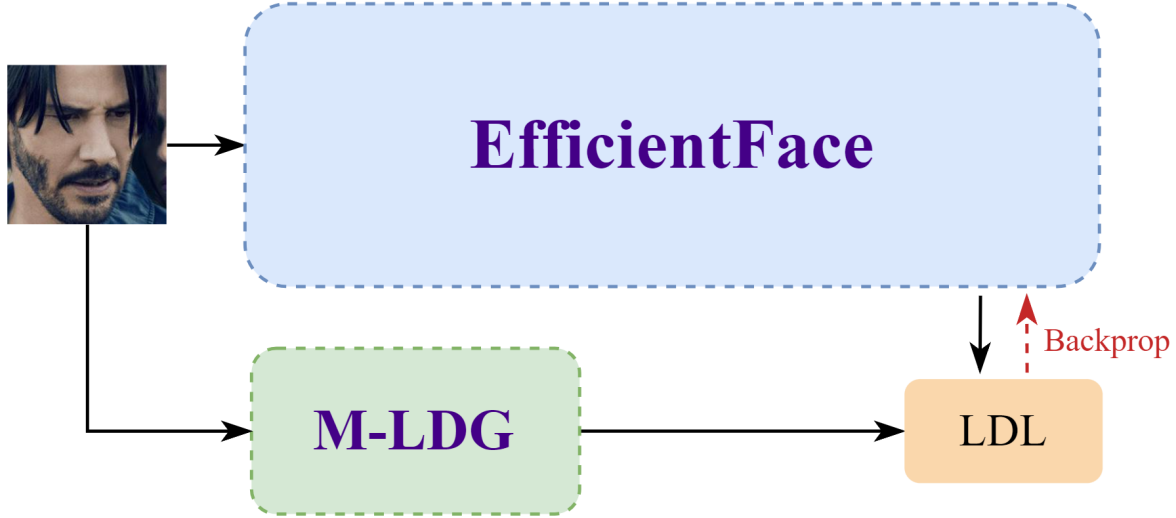


Figure 5.2: Training pipeline for the second phase of the proposed algorithm. The modified LDG is used to train EfficientFace on the AffectNet dataset.

5.4.2. Dataset preprocessing

Dataset preprocessing involves standardizing and transforming the facial images to create model-ready inputs. For each image, the pipeline first detects and aligns faces using RetinaFace [5] to account for pose variation and ensure facial landmarks are centered and consistently oriented. Images are then resized and padded to a fixed size of 112 x 112 pixels. These transformations ensure that all images fed to the models are properly aligned, uniformly sized, and consistently formatted, which is critical for robust model training. These steps are carried out once outside the training pipeline.

Additional preprocessing steps are applied at training and validation time. These steps follow the exact same procedure as the original EfficientFace architecture. For training, images are first randomly resized and cropped to the target size, followed by random horizontal flipping for data augmentation. Each image is then normalized using mean and standard deviation values matching the ImageNet dataset (mean=[0.485, 0.456, 0.406]; std=[0.229, 0.224, 0.225]). For validation and testing, images are simply resized to the target size, and normalized in the same way, but no random augmentation

is applied.

5.4.2.1. Image upscaling

An important implementation detail is that the APViT and modified LDG architectures expect and are trained on 112 x 112 pixel images as input, while EfficientFace expects 224 x 224 sized images. To address this, we upscale the images to 224 x 224 during training in the second phase of the proposed algorithm. The upscaling is carried out using bilinear interpolation.

5.4.3. Hyperparameter optimization

Both phases of the experimental design follow the same general procedure. First, the experiment is executed across all factor combinations. From these runs, the best-performing combinations are selected and subjected to hyperparameter optimization. The optimization searches over learning rate, momentum, weight decay, and the learning rate schedule (adjustment frequency and reduction factor), using a Bayesian search with early termination (Hyperband [18]) to select the best configuration. The resulting optimized model is taken as the winner or representative of that phase. Phase II uses the Phase I winner whenever the modified LDG is the selected guidance network: the best modified LDG from Phase I is loaded as the frozen label distribution generator for training EfficientFace in the corresponding Phase II condition.

5.4.4. Training APViT on AffectNet

As noted in the Requirements section, pretrained APViT weights for AffectNet were not publicly available at the time of this work. The network was therefore trained from scratch following the procedure described in the original APViT paper [36].

The public APViT codebase is oriented toward training on the RAF-DB dataset, so modifications were necessary to support training on AffectNet. These adaptations include changes to the data loading pipeline, class configuration, and evaluation scripts. The modified codebase is publicly available at <https://github.com/DavidCrgh/APViT>.

Table 5.1 lists the key training hyperparameters used for this run. These values were taken directly from the configuration file provided in the original repository and kept unchanged.

Table 5.1: APViT training hyperparameters used for training on AffectNet.

Hyperparameter	Value
Training iterations (batches)	6000
Input image size	112×112 px
Batch size	128
Optimizer	SGD
Learning rate	5×10^{-4}
Momentum	0.9
Weight decay	5×10^{-4}
LR schedule	Cosine Annealing
Gradient clip max norm	10.0
ViT depth	8 layers
ViT embedding dim	768
Attention heads	8
CNN pool (patches retained)	160

The trained APViT achieved 64.2% accuracy on AffectNet-7 and 57.31% on AffectNet-8. These results are below the 66.91% reported in the original paper [36], which was obtained on the same AffectNet-7 split, suggesting there is room for further tuning. Nevertheless, the trained model provides a functional teacher network for the proposed knowledge distillation pipeline.

5.4.5. Limitations and Requirements

This section outlines the key limitations and requirements of our proposed design, derived from the scope and limitations defined for this research project.

5.4.5.1. Limitations

Refer to section 3.3 for the limitations of this research project.

5.4.5.2. Requirements

Pretrained models: The proposed approach requires access to model weights for the APViT network pretrained on the AffectNet dataset. To the best of our knowledge, these weights are not publicly available. As such, we retrained the network from scratch using the same procedure as described in the original APViT paper [36].

Computational resources: The main computational resource required for executing the proposed approach is the available GPU memory. Phase I of the design requires more memory since it leverages the two heaviest networks: APViT and the modified LDG. In executions of Phase I, we noticed a memory requirement of approximately 16GB of VRAM per training run.

Dataset access: The execution of the proposed approach requires access to the 7-class and 8-class subsets of the AffectNet dataset [27].

Software environment: Execution of the training scripts requires an environment with the following software:

- Python 3.10
- PyTorch 1.11 and CUDA 11.8
- MNCV 1.7.1
- WandB

- RetinaFace [5]
- A copy of the main code repository. (See 5.4.6).

Face alignment tools: The main code repository contains a script for face alignment using RetinaFace. Any of its dependencies must be installed in the execution environment.

5.4.6. Implementation

The main code repository is hosted on GitHub and can be accessed at the following URL: <https://github.com/DavidCrgh/FER-MLDG>. The `src/networks` directory contains the implementations of the APViT, LDG, and EfficientFace architectures. Dataset preprocessing code, including face alignment with RetinaFace and conversion to the formats required by each network, is located in the `src/preproc` directory. Experiment execution is organized through Jupyter notebooks. The notebooks handle environment setup, configuration loading, and invocation of the underlying training scripts. Notebooks are also provided for executing the hyperparameter optimization process using WandB Sweeps.

Results

“The purpose of computing is
insight, not numbers.”

Richard Hamming

This chapter reports the findings of the two experimental phases described in chapters Methodology and Design. The response variable for both phases is top-1 validation accuracy on the AffectNet dataset [27]. We present factorial results, main effect estimates, learning curve observations, hyperparameter optimization outcomes, and statistical test results in the order they were produced. Interpretation and analysis of these findings are deferred to the Discussion chapter.

6.1. Phase I: label distribution generator ablation

6.1.1. Factorial results

The Phase I full 2^4 factorial design yielded 16 experimental runs covering all combinations of the four factors defined in Table 4.1: pretrained backbone weights (A), local-global feature extraction modules (B), APViT guidance (C), and dataset (D). The treatment combinations are enumerated in Table 4.3. Table 6.1 presents all 16 runs ranked by best validation accuracy.

The 16 runs span a range from 63.31% to 48.71%, with a mean of 56.13% and an overall standard deviation of 5.05%. The eight runs with APViT guidance enabled (C+) occupy ranks 1 through 8 and cluster between 56.76% and 63.31%, while the eight runs without guidance (C-) occupy ranks 9 through 16 and cluster between 48.71% and 55.51%. The highest accuracy (63.31%) was achieved by treatment 12 (C+, A-, B-, AffectNet-7) and the lowest (48.71%) by treatment 15 (C-, A-, B-, AffectNet-8). Within each C level, runs trained on AffectNet-7 consistently outperformed their AffectNet-8 counterparts by approximately 6 percentage points, confirming the dominant roles of factors C and D before formal effect estimation.

Rank	Treat.	A	B	C	D	Best Acc. (%)	Best Ep.
1	12	−	−	+	AN-7	63.31	17
2	0	+	+	+	AN-7	63.29	15
3	4	+	−	+	AN-7	63.20	13
4	8	−	+	+	AN-7	63.14	27
5	1	+	+	+	AN-8	57.06	14
6	5	+	−	+	AN-8	56.99	18
7	9	−	+	+	AN-8	56.84	29
8	13	−	−	+	AN-8	56.76	27
9	2	+	+	−	AN-7	55.51	23
10	14	−	−	−	AN-7	55.34	27
11	6	+	−	−	AN-7	55.26	28
12	10	−	+	−	AN-7	55.17	27
13	11	−	+	−	AN-8	49.24	24
14	7	+	−	−	AN-8	49.24	24
15	3	+	+	−	AN-8	49.01	29
16	15	−	−	−	AN-8	48.71	28

AN-7 = AffectNet-7; AN-8 = AffectNet-8. + / − denote the high / low factor level. Treatment IDs correspond to the run numbering defined in Table 4.3.

Table 6.1: All 16 treatment combinations of the Phase I 2^4 factorial design, ranked by best validation accuracy. Factors are: A (pretrained backbone weights), B (local-global feature extraction modules), C (APViT guidance), D (dataset subset).

6.1.2. Main effects and interactions

Main effect estimates were computed using Yates’s algorithm [28]. The results are summarized in Table 6.2 and visualized in figure 6.1.

Factor C (APViT guidance) produced the largest main effect at +7.889 percentage points (pp), followed by factor D (dataset) at -6.297 pp, indicating that the AffectNet-7 subset yields higher accuracy than AffectNet-8. Factors A (pretrained backbone weights) and B (local-global feature extraction modules) showed negligible effects of +0.129 pp and +0.056 pp, respectively. All interaction effects were also negligible: the C×D interaction was +0.025 pp, and all remaining two-factor and higher-order interactions fell below 0.18 pp in absolute value. The within-cell standard deviations, obtained after pooling A and B, ranged from 0.08% to 0.25%,

Factor / Interaction	Effect	Abs(Effect)
C (APViT guidance)	+7.889	7.889
D (Dataset)	−6.297	6.297
A (Pretrained backbone)	+0.129	0.129
A×D	−0.058	0.058
B×D	−0.056	0.056
B (GL modules)	+0.056	0.056
B×C	−0.040	0.040
C×D	+0.025	0.025
A×C	−0.010	0.010
A×B	−0.008	0.008
Higher-order (≥ 3 factor)	—	< 0.18

Table 6.2: Main effect and interaction estimates for the Phase I 2^4 factorial design computed with Yates’s algorithm. Values are in percentage points (pp). Effects are listed in decreasing order of absolute value.

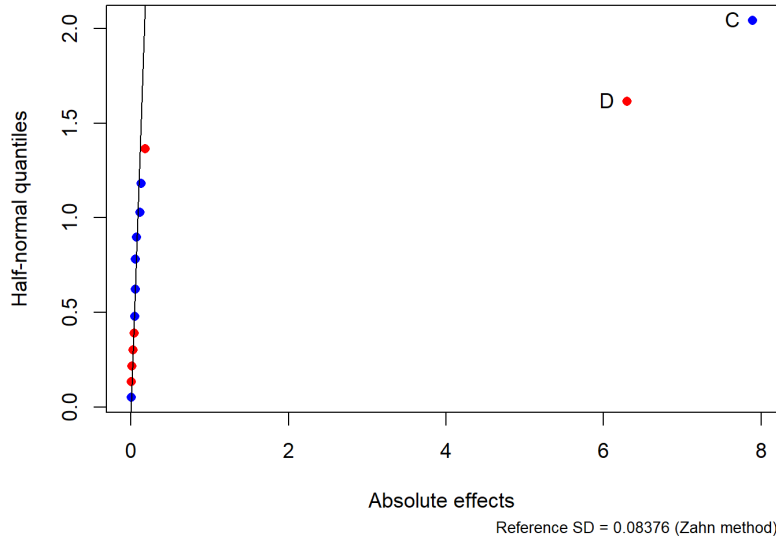


Figure 6.1: Half-normal probability plot of the Phase I effect estimates from Yates’s algorithm. Effects that deviate substantially from the reference line are considered active [28]. Only active effects are labeled.

confirming that A and B contribute no meaningful variance to the response.

The C×D interaction quantifies whether the benefit of APViT guidance varies across datasets. The near-zero interaction estimate of +0.025 pp indicates that the

contributions of C and D are negligible. Figure 6.2 shows the interaction line plot for C and D after pooling A and B.

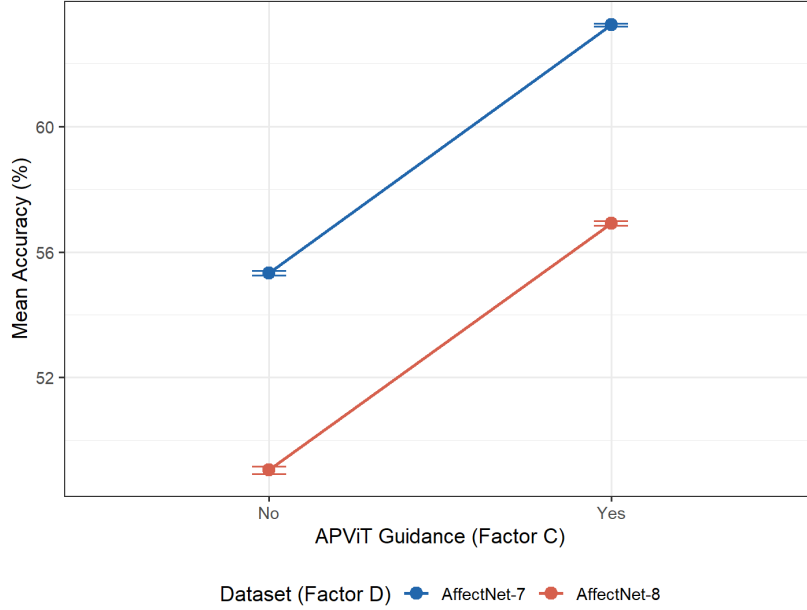


Figure 6.2: Interaction plot for factors C (APViT guidance) and D (dataset) after pooling negligible factors A and B. The near-parallel lines confirm the negligible $C \times D$ interaction (+0.025 pp).

6.1.3. Learning behavior

Training and validation accuracy curves were recorded for all 16 Phase I runs via WandB. Figure 6.3 presents representative curves for the four $C \times D$ conditions.

We observe two distinct training behaviors depending on the state of factor C. Runs with APViT guidance (C+) converged to their best validation accuracy between epochs 13 and 27, with a mean train-validation gap of approximately 5-7%. Runs without APViT guidance (C-) exhibited severe overfitting, with training accuracy reaching 75-76% while validation accuracy plateaued early, yielding a mean train-validation gap of approximately 24%. The learning rate schedule, which applies a decay factor at epochs 10 and 20, produced visible accuracy steps in C- runs that are attenuated or absent in

6.1. Phase I: label distribution generator ablation

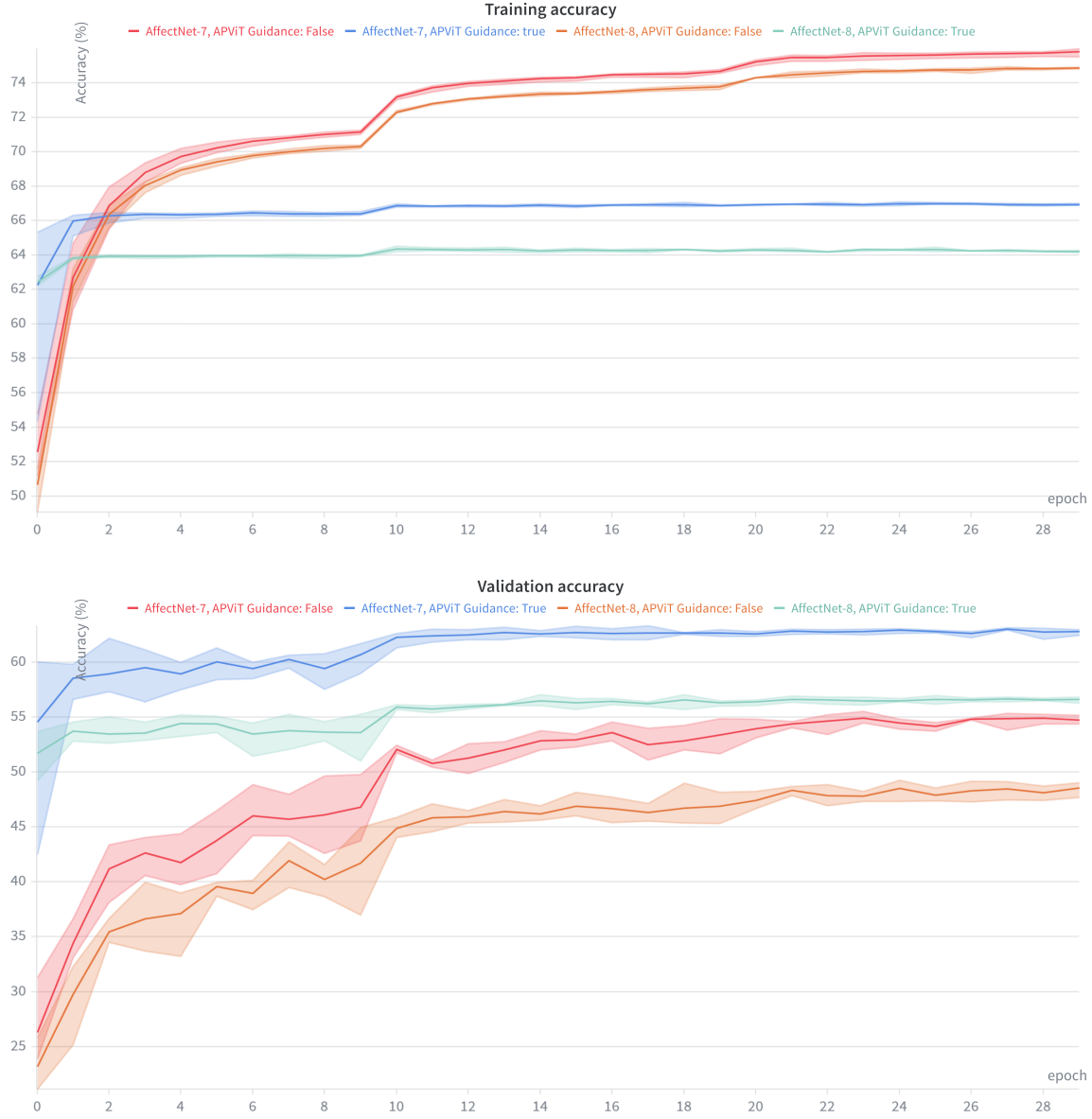


Figure 6.3: Representative training and validation accuracy curves for the four $C \times D$ conditions in Phase I. Each curve aggregates runs (mean) within the same $C \times D$ cell across varying A and B levels.

$C+$ runs. Across all conditions, runs trained on AffectNet-8 plateaued approximately 6 pp below their AffectNet-7 counterparts.

6.1.4. Hyperparameter optimization and the M-LDG

Following the factorial experiment, a Bayesian hyperparameter search using WandB Sweeps with Hyperband early termination [18] was applied to the two best-performing factor combinations for each dataset (0, 1, 5, and 12). Eight 12-epoch trials were conducted across the two datasets, after which the best-identified configurations were confirmed with full 30-epoch runs. Table 6.3 reports all eight sweep trials.

#	Dataset	LR	M	WD	F	Freq.	Best Acc. (%)
1	AN-7	0.00205	0.865	5.08×10^{-3}	0.561	12	64.77
2	AN-7	0.00205	0.944	1.35×10^{-6}	0.193	10	63.97
3	AN-7	0.01461	0.861	2.50×10^{-6}	0.138	10	63.89
4	AN-7	0.04980	0.888	8.01×10^{-6}	0.338	5	63.46
5	AN-7	0.02782	0.931	1.76×10^{-5}	0.600	8	63.20
6	AN-8	0.01856	0.932	1.16×10^{-6}	0.474	3	57.49
7	AN-8	0.02003	0.937	1.37×10^{-6}	0.516	3	57.29
8	AN-8	0.00430	0.885	4.09×10^{-6}	0.543	12	57.21

Table 6.3: Hyperparameter configurations and best validation accuracies for the eight Phase I sweep trials. Accuracy values are from 12-epoch runs with Hyperband early termination. M = Momentum, WD = Weight Decay, F = Factor, Freq = Frequency.

The resulting optimized network, achieved 64.34% on AffectNet-7 (best epoch: 9) and 57.71% on AffectNet-8 (best epoch: 12) in their respective 30-epoch confirmatory runs. These results represent improvements of +1.03 pp and +0.65 pp over the best ablation runs with C=True (63.31% and 57.06%). This optimized variant is used as the singular M-LDG network in the next phase of this study. Table 6.4 compares the default and optimized hyperparameter configurations used to train the M-LDG.

The optimized configurations differ from the defaults in three main respects: the learning rate is 3 to 4 times lower, the weight decay is 10 to 70 times lower, and the LR reduction factor is substantially less aggressive (0.209 for AffectNet-7 and 0.450

Parameter	Default (Ablation)	Optimized (AN-7)	Optimized (AN-8)	Change
Learning Rate	0.100	0.024	0.032	↓ 3–4× lower
Momentum	0.900	0.860	0.948	Dataset-specific
Weight Decay	1.0×10^{-4}	8.8×10^{-6}	1.5×10^{-6}	↓ 10–70× lower
LR Factor	0.100	0.209	0.450	↑ Less aggressive
Adjust Freq	10	10	3	Dataset-specific

Table 6.4: Default (ablation) and optimized hyperparameter configurations for the M-LDG on each AffectNet subset. Arrows indicate the direction of change relative to the default.

for AffectNet-8, compared to the default of 0.1). The AffectNet-7 M-LDG converged rapidly, reaching within 0.5 pp of its best accuracy by epoch 6, with a final train-validation gap of 2.71 pp. The AffectNet-8 M-LDG required more epochs to stabilize, with a final gap of 7.19 pp, reflecting the increased difficulty of the 8-class task.

6.2. Statistical assumption tests

Prior to the two-way ANOVA, we verified the normality of residuals and the homogeneity of variance as required by the ANOVA assumptions. Both assumption tests were conducted on the Phase I data after pooling the negligible factors A and B into the error term, retaining C and D as the active factors.

The Shapiro-Wilk test for normality of residuals yielded $W = 0.9447$ ($p = 0.410$), providing no evidence against the normality assumption. Levene’s test for homogeneity of variance produced $F(3, 12) = 1.317$ ($p = 0.314$), indicating no significant difference in variance across the four C×D cells. Both assumptions for parametric ANOVA are therefore satisfied. Figure 6.4 shows the normal Q-Q plot of residuals and the residuals-versus-fitted plot used to assess these assumptions visually.

6.3. Two-way ANOVA results

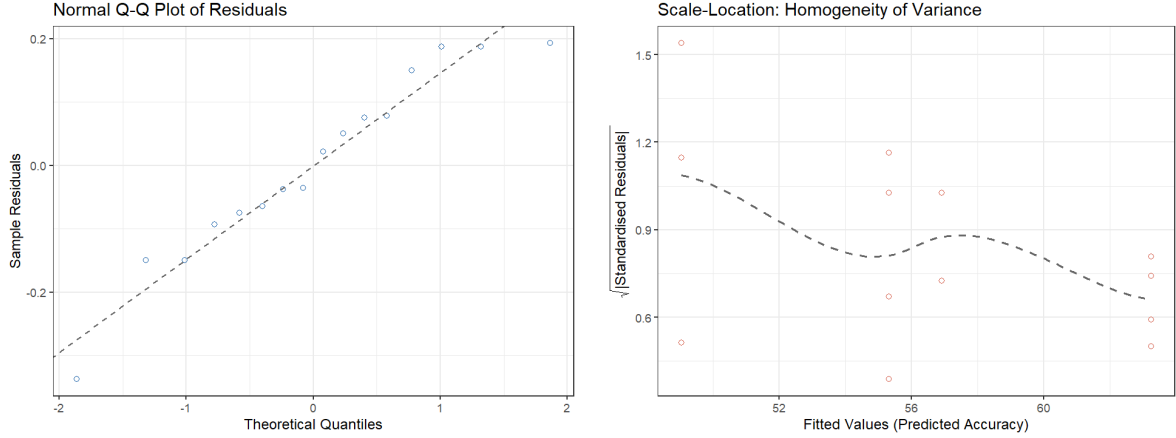


Figure 6.4: Phase I ANOVA assumption checks after pooling factors A and B. Left: normal Q-Q plot of residuals; close alignment with the reference diagonal supports normality (Shapiro-Wilk $W = 0.9447$, $p = 0.410$). Right: residuals versus fitted values for the two-way ANOVA model; relatively constant spread suggests homoscedasticity, confirmed by Levene’s test ($F(3, 12) = 1.317$, $p = 0.314$).

6.3. Two-way ANOVA results

A two-way ANOVA was performed on the Phase I data with best validation accuracy as the response and APViT guidance (C) and dataset (D) as the predictors, following the procedure defined in the Methodology chapter. Given their low effect estimates, we pool together factors A and B to increase the reliability of the results. The criteria for pooling is based on visual analysis via the half-normal plot in Figure 6.1. Table 6.5 presents the ANOVA table.

Source	df	SS	MS	F	p-value
APViT guidance (C)	1	248.97	248.97	9196.84	$< 2 \times 10^{-16}$ ***
Dataset (D)	1	158.59	158.59	5858.17	$< 2 \times 10^{-16}$ ***
$C \times D$	1	0.002	0.002	0.092	0.767
Residuals	12	0.325	0.027		

Significance codes: *** $p < 0.001$. Factors A and B were pooled into the residuals.

Table 6.5: Two-way ANOVA results for Phase I. Factors A and B were pooled into the residual after the effect estimation step; the model retains only factors C (APViT guidance) and D (dataset).

Both main effects were highly significant, while their interaction was not. APViT guidance (C) yielded $F(1, 12) = 9196.84$ ($p < 2 \times 10^{-16}$), the dataset (D) yielded $F(1, 12) = 5858.17$ ($p < 2 \times 10^{-16}$), and the C×D interaction was negligible ($F(1, 12) = 0.092$, $p = 0.767$). The residual term in Table 6.5 is very small: repeated runs under the same condition landed within roughly 0.16 percentage points of one another, so the observed differences between conditions are highly reproducible rather than the product of random fluctuation.

To measure how much each factor matters in practice, we report effect sizes as eta-squared (η^2), the proportion of the total variation in accuracy explained by a factor. Dividing each factor’s sum of squares in Table 6.5 by the total sum of squares (407.89) gives $\eta^2 = 0.610$ for APViT guidance and $\eta^2 = 0.389$ for the dataset, with the interaction below 0.001. Together these two factors explain about 99.9% of the systematic variation in accuracy. We therefore reject the null hypothesis that all treatment combinations yield the same mean accuracy: both APViT guidance and the choice of dataset produce statistically significant differences in performance.

6.4. Phase II: EfficientFace guidance comparison

6.4.1. Cell means and replication

Phase II employed a 2^2 factorial design with two replicates per treatment combination, as described in Table 4.4. The two factors were guidance network (G: M-LDG or APViT) and dataset (D: AffectNet-7 or AffectNet-8). A total of eight EfficientFace training runs were conducted, each for 45 epochs. Table 6.6 reports the individual run results and cell means.

The M-LDG-guided conditions achieved cell means of 62.71% on AffectNet-7 (SD

Guidance (G)	Dataset (D)	Rep. 1	Rep. 2	Cell Mean	SD	n
APViT	AffectNet-7	62.31	61.89	62.10	0.303	2
APViT	AffectNet-8	56.11	55.49	55.80	0.442	2
M-LDG	AffectNet-7	63.00	62.43	62.71	0.404	2
M-LDG	AffectNet-8	55.99	55.81	55.90	0.124	2

Table 6.6: Phase II best validation accuracy (%) for each run, cell means, and within-cell standard deviations. Each cell contains two replicates of EfficientFace trained with the specified guidance network on the specified dataset.

= 0.404%) and 55.90% on AffectNet-8 (SD = 0.124%). The APViT-guided conditions achieved 62.10% (SD = 0.303%) and 55.80% (SD = 0.442%), respectively.

Figure 6.5 shows the cell means with error bars.

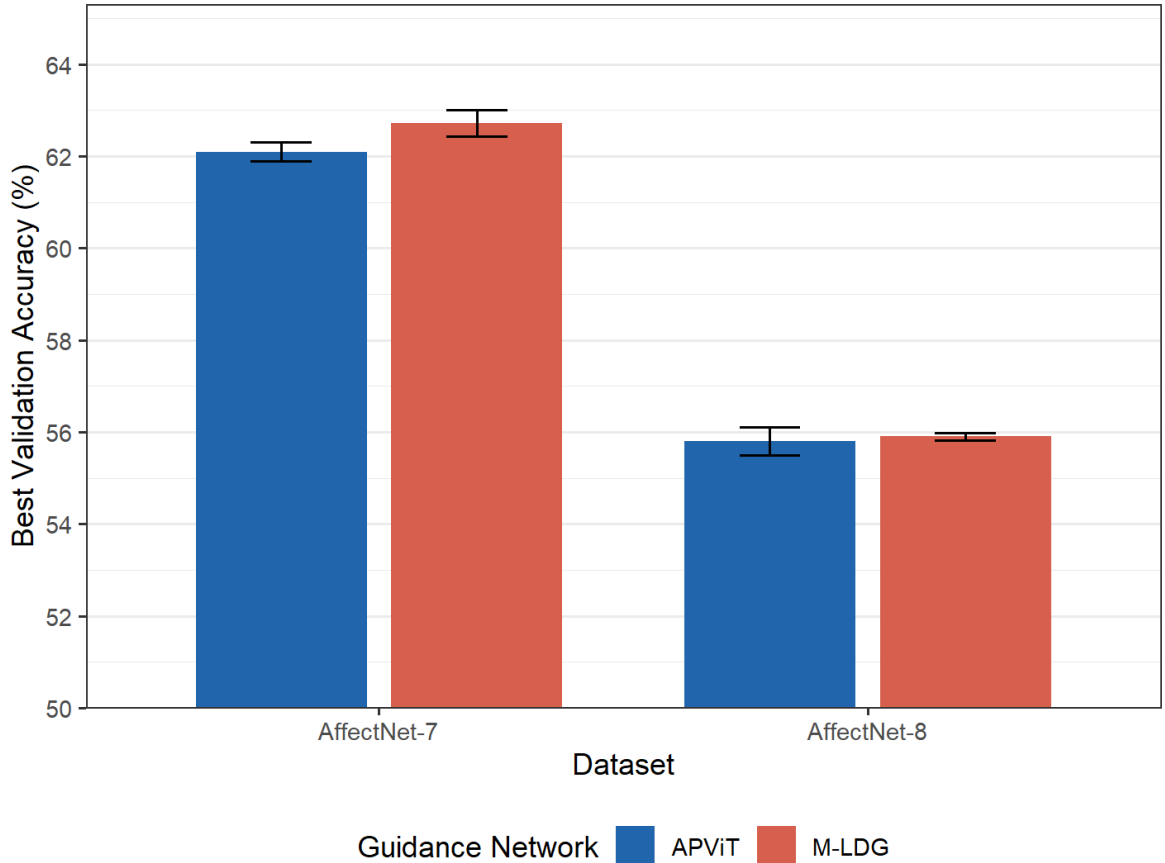


Figure 6.5: Cell means for EfficientFace best validation accuracy under each guidance network and dataset condition in Phase II. Error bars indicate the range between the two replicates in each cell.

6.4.2. Significance testing

Welch’s t-tests were applied separately for each dataset to compare the mean accuracy of M-LDG-guided and APViT-guided EfficientFace, as specified in the Methodology chapter.

For AffectNet-7, the mean accuracy was 62.71% for M-LDG (SD = 0.404%) and 62.10% for APViT (SD = 0.303%). The observed difference was +0.614 pp in favor of the M-LDG. The Welch t-test yielded $t(1.85) = -1.720$ with $p = 0.237$ and a 95% confidence interval of $[-1.044, 2.272]$ pp. For AffectNet-8, the mean accuracy was 55.90% for M-LDG (SD = 0.124%) and 55.80% for APViT (SD = 0.442%). The observed difference was +0.100 pp in favor of the M-LDG. The Welch t-test yielded $t(1.16) = -0.308$ with $p = 0.804$ and a 95% confidence interval of $[-2.917, 3.117]$ pp.

Neither comparison reached statistical significance. The wide confidence intervals spanning zero in both cases indicate that the direction of the difference cannot be established at the current sample size. With $n = 2$ replicates per cell, the Phase II tests are severely underpowered and can only detect very large effects; the failure to reach significance should be interpreted as inconclusive rather than as evidence of equivalence between M-LDG and APViT as guidance networks for EfficientFace.

6.4.3. Hyperparameter optimization

Following the same Bayesian sweep procedure as Phase I, hyperparameter optimization was applied to two Phase II conditions: M-LDG with AffectNet-7 and APViT with AffectNet-8. Five 20-epoch sweep trials were conducted per combination, followed by 65-epoch confirmatory runs using the best identified configuration.

The optimized confirmatory runs achieved 62.29% for M-LDG on AffectNet-7 and

55.01% for APViT on AffectNet-8. These are lower than the corresponding Phase II baseline runs conducted with default hyperparameters, which achieved 63% and 56.11%, respectively. The differences represent decreases of -0.43 pp and -0.45 pp. Hyperparameter optimization in Phase II therefore produced a negative result: the default training configuration (learning rate 0.1, momentum 0.9, weight decay 10^{-4} , LR factor 0.1, adjustment frequency 10) outperformed all optimized configurations identified by the sweep.

6.5. Additional obtained metrics

6.5.1. Cross-phase accuracy progression

To summarize the accuracy gains at each stage of the proposed pipeline, Table 6.7 presents the cross-phase accuracy progression from the unguided LDG baseline through to the final EfficientFace variants. This table directly addresses specific objective 3 from the Hypothesis and Objectives chapter, which calls for training two EfficientFace variants and studying their performance relative to the baseline and to each other.

The unguided M-LDG baseline ($C=False$) achieved 55.51% on AffectNet-7 and 49.24% on AffectNet-8. Adding APViT knowledge distillation ($C=True$) raised accuracy to 63.31% and 57.06%, respectively. The M-LDG, after hyperparameter optimization, reached 64.34% (AffectNet-7) and 57.71% (AffectNet-8). EfficientFace trained with M-LDG-generated soft labels achieved 63.00% (AffectNet-7) and 55.99% (AffectNet-8), while EfficientFace trained with APViT-generated soft labels achieved 62.31% and 56.11%. For reference, the original EfficientFace authors report 63.7% on AffectNet [40] and the APViT teacher used in this study reports 66.91% on AffectNet [36].

Stage	AffectNet-7	AffectNet-8
M-LDG baseline (C=False, ablation)	55.51	49.24
M-LDG + APViT guidance (C=True, ablation)	63.31	57.06
M-LDG (Phase I optimized)	64.34	57.71
EfficientFace guided by M-LDG (Phase II)	63.00	55.99
EfficientFace guided by APViT (Phase II)	62.31	56.11
Published EfficientFace [40]	63.70	59.89
Published APViT [36]	66.91	—
APViT trained from scratch	64.2	57.31

Table 6.7: Accuracy (%) at each stage of the proposed knowledge transfer pipeline for both AffectNet subsets. Published baselines from [40] and [36] are provided for reference. We also present the performance of our own APViT model trained from scratch on AffectNet-7 and AffectNet-8.

6.5.2. Per-class behavior

In addition to the top-1 accuracy response variable, per-class validation confusion matrices were logged for all runs via WandB. We report here the confusion matrices for the best M-LDG-guided EfficientFace variants obtained in Phase II: the AffectNet-7 model (63% best accuracy) and the AffectNet-8 model (55.99% best accuracy). These matrices capture the distribution of correct and incorrect predictions across expression classes and complement the main accuracy reported throughout this chapter.

Figure 6.6 shows the normalized confusion matrix for the best M-LDG-guided EfficientFace on AffectNet-7 and figure 6.7 shows the corresponding matrix for AffectNet-8.

6.5.3. Per-class and macro-averaged metrics

Beyond the top-1 accuracy response variable, we computed per-class and macro-averaged classification metrics for the five pipeline stages of table 6.7 on both AffectNet subsets, derived from the same logged confusion matrices used in section

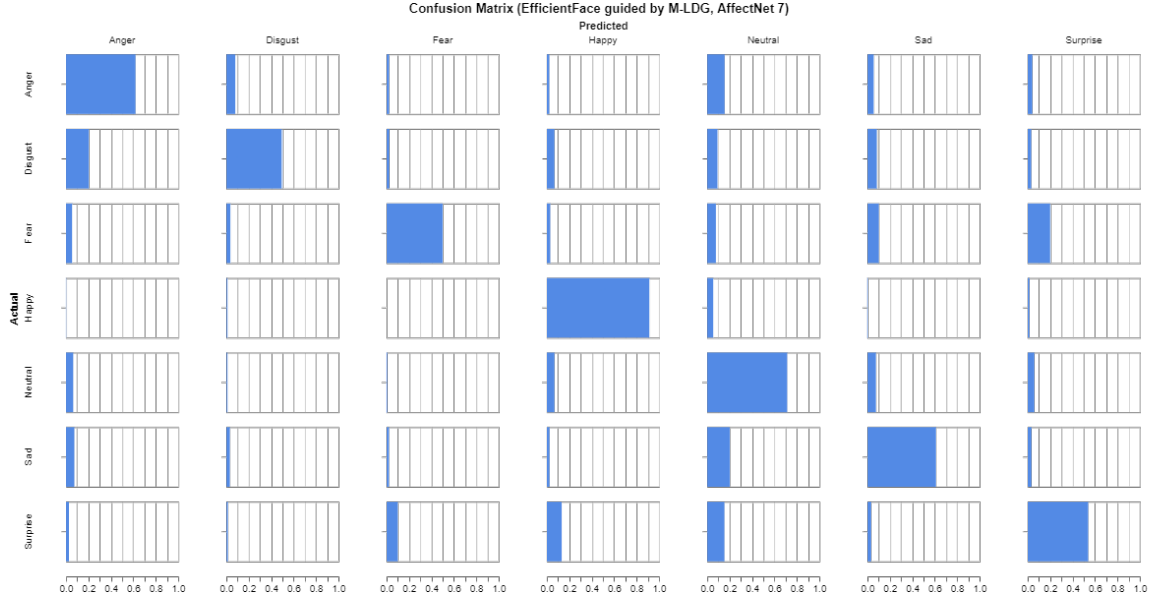


Figure 6.6: Normalized confusion matrix for the best M-LDG-guided EfficientFace variant on AffectNet-7. Rows represent true classes; columns represent predicted classes. Values indicate the fraction of samples in each true-class row assigned to each predicted class.



Figure 6.7: Normalized confusion matrix for the best M-LDG-guided EfficientFace variant on AffectNet-8, including the contempt class. Rows and columns follow the same convention as figure 6.6.

6.5.2. These metrics were not subjected to the formal statistical analysis applied to top-1 accuracy in Phases I and II and are reported here as descriptive summaries only.

6.5. Additional obtained metrics

Table 6.8 reports per-class recall, and table 6.9 reports the macro-averaged precision, recall, and F1 score alongside top-1 accuracy for all ten trained models.

AffectNet-7					
Emotion	M-LDG base	M-LDG+APViT	M-LDG optim.	EF+APViT	EF+M-LDG
Anger	0.562	0.660	0.654	0.656	0.622
Disgust	0.282	0.450	0.514	0.436	0.506
Fear	0.440	0.540	0.538	0.498	0.520
Happy	0.936	0.914	0.908	0.928	0.914
Neutral	0.784	0.726	0.716	0.700	0.704
Sad	0.518	0.642	0.634	0.622	0.618
Surprise	0.364	0.500	0.540	0.522	0.526

AffectNet-8					
Emotion	M-LDG base	M-LDG+APViT	M-LDG optim.	EF+APViT	EF+M-LDG
Anger	0.564	0.638	0.642	0.642	0.612
Contempt	0.004	0.134	0.186	0.124	0.164
Disgust	0.286	0.460	0.466	0.450	0.452
Fear	0.444	0.504	0.522	0.502	0.478
Happy	0.950	0.920	0.894	0.908	0.898
Neutral	0.776	0.682	0.714	0.684	0.714
Sad	0.566	0.702	0.682	0.686	0.666
Surprise	0.348	0.524	0.510	0.492	0.494

Table 6.8: Per-class recall for the five pipeline stages of table 6.7 on both AffectNet subsets.

AffectNet-7					
Metric	M-LDG base	M-LDG+APViT	M-LDG optim.	EF+APViT	EF+M-LDG
Top-1 Acc. (%)	55.51	63.31	64.34	62.31	63.00
Precision	0.626	0.653	0.658	0.647	0.643
Recall	0.555	0.633	0.643	0.623	0.630
F1	0.543	0.629	0.641	0.618	0.627

AffectNet-8					
Metric	M-LDG base	M-LDG+APViT	M-LDG optim.	EF+APViT	EF+M-LDG
Top-1 Acc. (%)	49.24	57.06	57.71	56.11	55.99
Precision	0.641	0.634	0.628	0.625	0.622
Recall	0.492	0.571	0.577	0.561	0.560
F1	0.454	0.549	0.562	0.540	0.543

Table 6.9: Macro-averaged precision, recall, and F1 score with the corresponding top-1 validation accuracy (%) for the five pipeline stages of table 6.7 on both AffectNet subsets. Macro averages weight every class equally regardless of its sample count.

6.5.4. Model complexity

Table 6.10 reports the parameter and MFLOPs (floating-point operations) counts for the four architectures involved in the proposed pipeline alongside the best top-1 validation accuracy each achieved on AffectNet-7. Counts were computed using the `thop` library [43], with each multiply-accumulate (MAC) operation counted as one FLOP. The 7-class and 8-class variants of each model differ only in their final classification head, so the 7-class figures serve as representative values.

Model	Best Acc. (%)	Params (M)	MFLOPs
M-LDG	64.34	56.43	6334.4
EfficientFace (Phase II)	63.00	1.28	158.6
Original LDG [40]	—	23.52	4131.7
APViT [36]	66.91	65.06	12400.1

Table 6.10: Best AffectNet-7 accuracy, parameter count (in millions), and MFLOPs for the four architecture families under study. M-LDG variants share identical complexity, as do all EfficientFace variants. Original LDG accuracy on AffectNet is not reported in [40].

6.5.5. Interpretability

To provide evidence of the spatial attention patterns learned by the main networks in our study, we apply class activation maps (CAMs) [41]. CAM visualizations highlight the image regions that contributed most strongly to the network’s predicted class, offering a spatial perspective on model behavior that complements the quantitative accuracy metrics reported in the preceding sections.

Figure 6.8 presents a grid of CAM visualizations for four of the five networks presented in the cross phase comparison (see table 6.7).

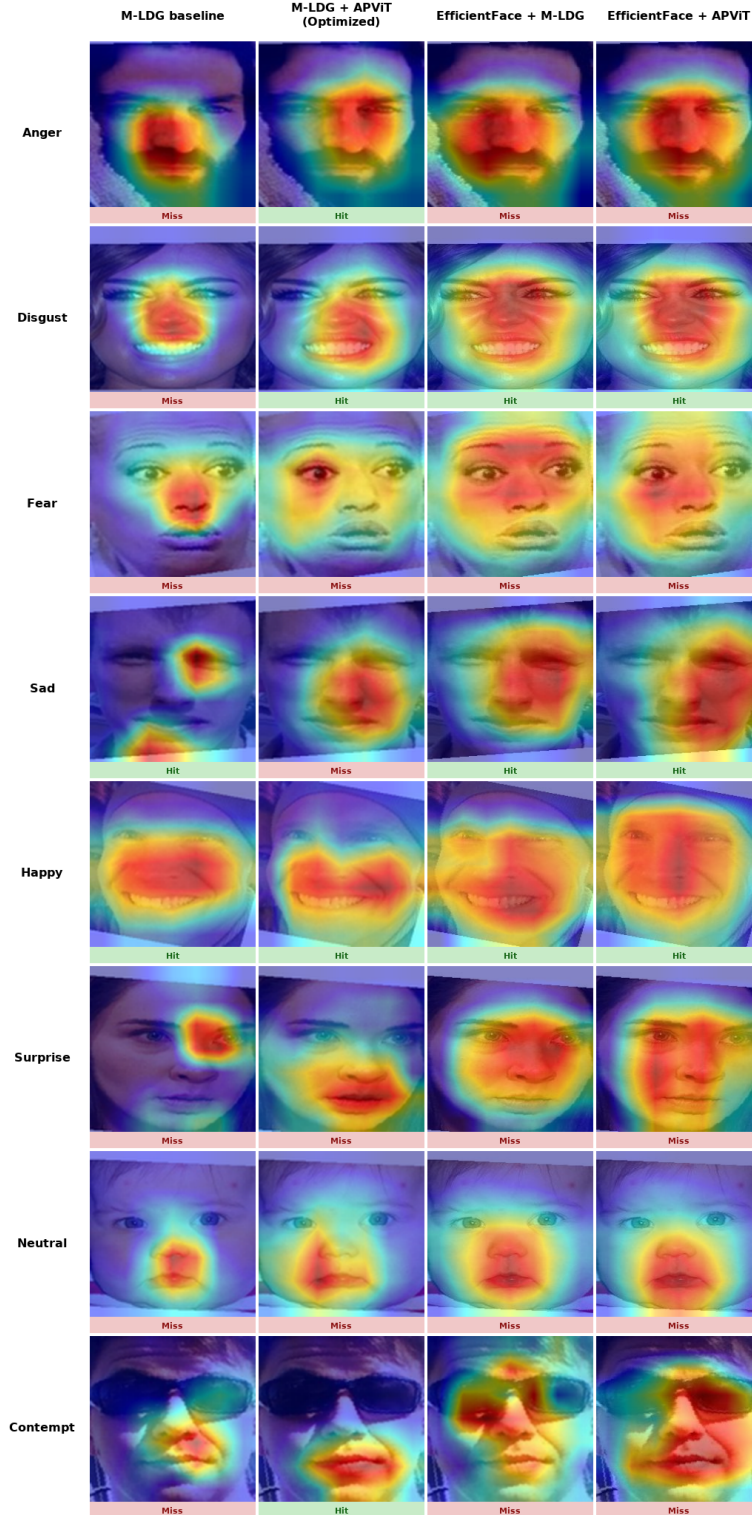


Figure 6.8: Class activation map (CAM) visualizations for four of the five main networks used in the cross phase comparison (see table 6.7). Each panel shows one validation sample with the model’s spatial attention overlaid. Warmer colors indicate higher activation. Hit and miss labels under each image indicate if the sample is correctly or incorrectly classified.

Discussion

“To consult the statistician after an experiment is finished is often merely to ask him to conduct a post mortem examination. He can perhaps say what the experiment died of.”

Ronald A. Fisher

This chapter presents the analysis and discussion of the findings reported in the previous chapter. We interpret the Phase I ablation and Phase II experimental results in light of the hypothesis and specific objectives defined in the Hypothesis and Objectives chapter, following the statistical procedures described in the Methodology chapter. The discussion is organized around five themes: the rejection of the main hypothesis, the mechanics of the two-hop knowledge transfer chain, the reasons why the individual LDG enhancements failed to propagate to the student network, a set of auxiliary observations, and a final synthesis that integrates these threads.

7.1. Revisiting the hypothesis

The main hypothesis of our study reads: *“Training EfficientFace with a label distribution generator enhanced with one or more of: FER pretrained backbone weights, local-global feature extraction modules, and/or vision transformer guidance, should increase its facial expression recognition accuracy on the AffectNet dataset when compared to the original baseline.”*

The evidence does not support this claim. The best-performing enhanced configuration, EF + M-LDG (EfficientFace trained with M-LDG-generated soft labels), achieved 63.00% on AffectNet-7 (AN-7) and 55.99% on AffectNet-8 (AN-8), while the published EfficientFace baseline [40] reports 63.70% and 59.89% on the same subsets. The alternative enhanced configuration, EF + APViT, similarly achieved 62.31% (AN-7) and 56.11% (AN-8), remaining below the published reference. Across both AffectNet subsets and both EfficientFace configurations tracked in Table 6.6, accuracy remained below the published reference. **We therefore reject the hypothesis.** The following sections will explore the reasons why this happened.

7.2. The two-hop knowledge transfer chain

This section examines the two hops of the knowledge transfer chain and identifies the possible causes responsible for the asymmetric outcomes observed across hops.

7.2.1. Asymmetric transfer across the two hops

The transfer chain runs from APViT (teacher, 65.06M params) through the modified label distribution generator (M-LDG, intermediary, 56.43M params) to EfficientFace (student, 1.28M params). The first hop from APViT to M-LDG proved highly effective: APViT itself achieves 64.2% and 57.31% on AffectNet-7 and AffectNet-8 respectively; the APViT-guided M-LDG achieves 64.34% on AffectNet-7, and 57.71% on AffectNet-8 (see Table 6.7). This means that in the first phase the student network was able to maintain the teacher’s performance while being around 13% smaller in terms of parameters. The second hop, from M-LDG to EfficientFace via label distribution learning, did not exhibit the same improvement.

EfficientFace trained with M-LDG-generated distributions (EF + M-LDG) achieved 63% and 55.99% on AffectNet-7 and AffectNet-8 respectively, drops of 1.34 pp and 1.81 pp below the M-LDG’s own accuracy. This is in direct contrast to the original EfficientFace result, where the label distribution learning-trained (LDL) student surpassed its LDG teacher [40]. That observation established the expectation that LDL produces a beneficial effect sufficient to overcome the student’s capacity disadvantage. In the present phase II setup, this expectation was not met.

We attribute the failure to a pronounced difference in capacity between the M-LDG and EfficientFace. The original EfficientFace paper [40] employed a LDG of 23.52M

parameters, a scale far closer to EfficientFace’s own parameter count. The M-LDG, at 56.43M parameters and 6334.4 MFLOPs, produces a ratio of approximately 44 to 1 relative to EfficientFace’s 1.28M parameters and 158.6 MFLOPs (Table 6.10). This expanded capacity gap is a key structural difference between the present setup and the original, and we argue it is responsible for the transfer asymmetry. The study by Mirzadeh et al. [25] on knowledge distillation via intermediate network assistance supports this claim as well. In their study, the authors found that large gaps in student and teacher sizes can significantly degrade performance.

7.2.2. Teacher-agnosticism of EfficientFace

Phase II found no statistically significant difference between M-LDG and APViT guidance on EfficientFace outcomes (Welch t-test $p = 0.237$ on AffectNet-7 and $p = 0.804$ on AffectNet-8), with 95% confidence intervals spanning zero on both subsets (Table 6.6; section 6.4.2). The G factor difference was a modest +0.357 pp in favor of M-LDG, and the two conditions differed by less than one percentage point on both datasets, with per-class recall and F1 gaps of at most 0.03 (Table 6.8). This result is best read as inconclusive rather than as evidence of equivalence: the comparison rested on $n = 2$ replications per cell, leaving the test statistically underpowered.

This result carries a double-edged interpretation. One positive reading of the result is that a computationally lighter intermediary can replace the full APViT transformer as the teacher signal during EfficientFace training without producing a meaningfully inferior student network, preserving the training-pipeline efficiency advantage that comes from avoiding the heavier APViT teacher. The negative reading, however, is that the higher performance of the M-LDG at the teacher level is downstream-irrelevant: an improvement of 7.889 pp in teacher accuracy failed to

propagate into statistically distinguishable gains in the student network, so the training cost incurred by enhancements to the M-LDG yield no visible benefit in the final deployed model.

A possible explanation for this outcome centers on the relative sizes of the three networks. M-LDG and APViT are comparable in scale to each other, yet both are substantially larger than EfficientFace (see Table 6.10). If the capacity gap between each teacher and the student is the dominant constraint on knowledge transfer, then the comparatively small size difference between the two teachers becomes secondary: neither network is well-positioned to bridge the gap into EfficientFace effectively, and substituting one for the other leaves the student’s learned representations essentially unchanged. This explanation is inferential, but it is consistent with the near-identical per-class profiles observed in Table 6.8.

7.3. Effectiveness of the LDG enhancements

This section examines why enhancements A (pretrained backbone weights) and B (local-global feature extraction modules) did not translate into improved M-LDG accuracy; the dominant role of enhancement C is analyzed in section 7.3.2. The focus here is on the internal training dynamics of the M-LDG rather than the downstream knowledge transfer chain analyzed in the previous section.

7.3.1. Negligible effects of factors A and B

The phase I ablation study confirms factor C as the only statistically significant booster of M-LDG accuracy in this setup (Table 6.5). Before drawing conclusions, we note that the standard deviation across all conditions ranged from 0.08 to 0.25 pp (Table 6.2), indicating that the experimental measurements are precise. The negligible

effects of A and B are therefore reliable findings rather than measurement artifacts obscured by noise.

We attribute the negligible effect of factor B to a structural mismatch between the origin of the local-global modules and the network into which they are inserted. The GL modules were originally designed for EfficientFace, a more shallow network whose reduced depth makes explicit local-global feature extraction a meaningful architectural addition [40]. Transplanting them into a ResNet-50 backbone places them inside a network that is already deep enough to extract these same representations across its numerous layers. We therefore argue that a mechanism tailored for a shallow network loses its effectiveness when embedded in a substantially deeper one. Nonetheless, a visual comparison of the activation patterns (see Figure 6.8) suggests that in spite of its lower accuracy, the EfficientFace variants that use GL modules do a better job of drawing attention towards the correct regions than the M-LDG variants that do not. There is one caveat regarding this interpretation for the contempt class, which will be discussed in section 7.4.

The negligible effect of factor A is best understood by examining which parts of the M-LDG network actually receive FER-pretrained initialization. The APViT-derived weights cover only the first three stages of the IR50 backbone [4]; the fourth stage and the classification head are not pretrained on FER and are instead initialized with weights from the face recognition pretraining on MS-Celeb-1M [11]. The richer, FER-specific knowledge encoded in the early stages therefore does not extend to the network’s higher-level layers. Because the upper stage must still learn high-level FER representations almost from scratch, the benefit of low-level FER-pretrained initialization is insufficient to guide learning at the levels where it matters most.

7.3.2. Evidence for APViT’s guidance and regularization role

Our ablation study provides strong evidence in favor of APViT’s effectiveness as a teacher network. Its presence improved M-LDG label distribution quality by +7.889 pp (see Table 6.5), a large and highly significant effect. Following hyperparameter optimization, the M-LDG reached 64.34% on AffectNet-7 and 57.71% on AffectNet-8, an improvement over the baseline of +8.83 pp and +8.47 pp respectively (Table 6.7).

A secondary observation can be made on the effect of APViT guidance on overfitting. As documented in section 6.1.3 and illustrated in Figure 6.3, M-LDG configurations without APViT guidance (C−) exhibited severe overfitting. The observed pattern places these configurations at high levels of training accuracy but low levels of validation accuracy. Whereas configurations with APViT guidance (C+) exhibit the opposite pattern, with higher validation accuracy and lower training accuracy. This reduction reveals a dual role for the label distribution learning (LDL) loss. Its primary role is what motivated its inclusion: the LDL loss causes the M-LDG to produce soft label distributions that encode the teacher’s uncertainty across emotion classes. Its secondary role, however, is that of an implicit regularizer: by constraining the M-LDG’s outputs to remain close to APViT’s more nuanced predictions, the LDL loss substantially reduces overfitting and steers the model toward a more generalizable function.

7.4. Auxiliary observations

This section provides complementary observations to the main discussion points exposed above. While not central to our study, they provide additional insights that could serve as a starting point for future work.

7.4.1. Challenge of the contempt class

Factor D (AffectNet-8 versus AffectNet-7) imposed an accuracy penalty of -6.297 pp across all M-LDG configurations (Table 6.2), reflecting the added difficulty of the eight-class subset. Contempt recall collapsed to 0.004 for the LDG baseline and recovered to only 0.186 for the optimized M-LDG (Table 6.8), indicating that this class remains nearly undetectable across all configurations. Contempt is the least frequent class in AffectNet-8 and also among the most challenging to label reliably, and its near-zero recall is likely one of the causes of the large gap between our best EfficientFace result (55.99%) and the published AffectNet-8 baseline (59.89%).

On a similarly interesting note, we have some light evidence for the contempt class being predicted by the mouth region. From the employed models, we note that the optimized M-LDG is slightly better than the EfficientFace at identifying contempt (M-LDG recall 0.186 vs EfficientFace recall 0.164). We complement this observation with the activation maps for the contempt class. The single sample shown in Figure 6.8 shows the M-LDG focusing on the mouth region to correctly classify the image. Analyzing a wider set of samples classified by the M-LDG we observe that correctly labeled images tend to focus almost exclusively on the mouth region, whereas incorrectly classified images tend to focus on other regions. This is shown in Figure 7.1. This observation should not be taken as definitive given the lack of backing quantitative evidence. Still, it is an interesting pattern that could be further explored in another study.

7.4.2. Macro-averaged precision, recall, and F1 patterns

Table 6.9 extends Table 6.7 with macro-averaged precision, recall, and F1 across all five configurations on AffectNet-7 and AffectNet-8. On AffectNet-7, recall and F1 mirror the top-1 accuracy: M-LDG base achieves recall 0.555 and F1 0.543; the optimized M-LDG peaks at 0.643 and 0.641. Precision is comparatively stable across

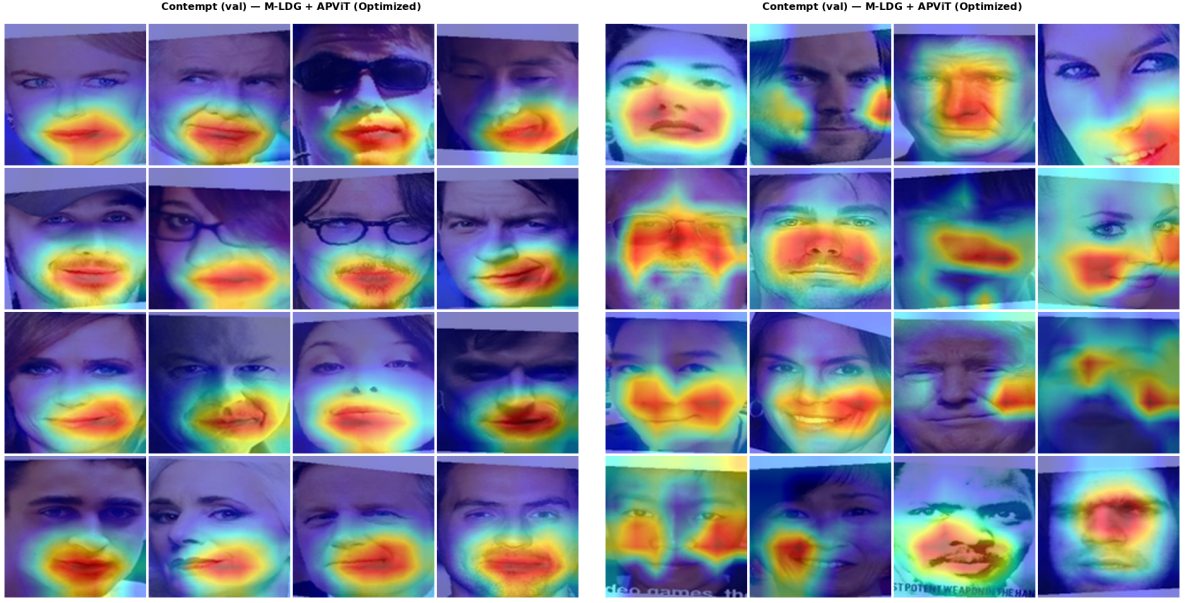


Figure 7.1: Activation maps for the contempt class on samples classified by the optimized M-LDG. Left: correctly classified samples; right: incorrectly classified samples.

configurations (0.626 to 0.658). This data suggests that the progression in Table 6.7 is not an artifact of accuracy as a metric choice.

On AffectNet-8, the M-LDG base records the highest precision (0.641) despite the lowest recall (0.492). We attribute this inversion to contempt-class suppression (section 7.4.1), i.e. rarely predicting a class yields few false positives for it, which inflates its precision contribution to the macro average. As APViT guidance broadens the distribution of predictions, false positives accumulate in suppressed categories, pulling precision to 0.628 for the optimized M-LDG while recall rises by 0.085. Precision alone is therefore a misleading indicator in the imbalanced AffectNet-8 setting.

The near-identical metric profiles of EF+APViT and EF+M-LDG corroborate the teacher-agnosticism finding from section 7.2.2. The largest single-metric gap between the two across both datasets is 0.009 (AffectNet-7 F1: 0.618 vs. 0.627; Table 6.9). On

AffectNet-8, all three measures differ by at most 0.003. These margins are consistent with the phase II results and reinforce the interpretation that substituting one teacher for the other leaves the student’s learned representations nearly indistinguishable.

7.4.3. Implementation constraints and their downstream effects

Because publicly available AffectNet-pretrained weights for APViT [36] did not exist at the time of this study, the network was trained from scratch on AffectNet, as described in chapter Design. This model did not reach the same level of performance as the published one, and we argue this likely introduced a performance ceiling for APViT in its role as teacher network. The downstream consequences of this ceiling extend across most experimental conditions: it affects all M-LDG variants trained with APViT guidance or FER pre-trained weights (factors A and C), it indirectly affects EfficientFace variants trained with M-LDG-generated distributions, and it affects the EfficientFace variant trained with APViT guidance directly.

We are careful to limit this claim. The from-scratch training constraint defines a possible ceiling of the pipeline, but it does not displace the capacity bottleneck as the likely primary culprit of the transfer failure. The from-scratch limitation is therefore best understood as a contributing factor that bounds how high the pipeline’s ceiling can reach, rather than as the root cause of why the student did not exceed the published baseline.

We attribute the lower accuracy of the in-house trained APViT to the difficulty in replicating the exact training procedure used in the original paper. The public APViT codebase [36] was oriented toward training on RAF-DB, and adapting it to AffectNet required non-trivial modifications to data loading, preprocessing, and training procedures. The EfficientFace codebase [40] is similarly RAF-DB-oriented by default,

and its adaptation is a likely source of noise that is hard to design for. These modifications introduce sources of experimental uncertainty that are difficult to fully quantify, and they provide additional context for interpreting the reproducibility gap with the published baseline discussed in section 7.1.

7.4.4. Hyperparameter optimization failure in Phase II

Despite using the same set of hyperparameters and levels as Phase I, the procedure did not yield any improvement over the default hyperparameters in Phase II. On the contrary, the attempted optimized runs performed approximately as well as the default configuration. This result is consistent with the findings of the Phase II comparison runs, which also did not yield any improvement over the default hyperparameters.

A likely explanation for this outcome is that the wide gap between teacher and student network capacities mentioned before has a performance-limiting effect. This means that even with varying sets of hyperparameters, the performance of the student network is capped at approximately the same level that was reached by the initial Phase II models. Further exploration of the hyperparameter space fails to yield any improvement due to it being constrained by the capacity gap. In all likelihood, the default configuration was already close to the optimal one for this setup only because of this limitation.

7.5. Synthesis

The central takeaway of this study is that enhancing the intermediary teacher worked, but that success did not carry through to the final student network. APViT guidance lifted the M-LDG to 64.34% on AffectNet-7 and 57.71% on AffectNet-8,

providing a substantial gain driven not just by richer soft labels, but by the implicit regularization effect that kept C- configurations from overfitting badly. The pipeline, at that level behaved as intended.

However, it was unable to overcome the scale mismatch at the second hop. A capacity ratio of roughly 44 to 1 between the M-LDG and EfficientFace is too wide for label distribution learning to bridge. The original EfficientFace paper demonstrated that a student can surpass its teacher through LDL, but that result assumed a far more modest gap between the two networks. Stretching that gap to the degree present here appears to neutralize whatever advantage the richer soft labels provide, and the near-identical outcomes of EF+M-LDG and EF+APViT reinforce the point: with a large scale difference, swapping one teacher for another does not make a significant difference in the student.

FER-pretrained weights and GL modules (factors A and B) tell a related story about transplanting mechanisms outside their intended context. When embedding the GL modules designed for the shallow EfficientFace into a deep ResNet-50, their contribution becomes redundant. FER-pretrained initialization only reached the first three backbone stages, leaving the upper layers, where high-level expression representations form, without any FER-specific grounding. Neither enhancement was wrong in principle; both were simply mismatched to the architecture they were applied to.

The from-scratch APViT training constraint sits on top of all of this as an additional limiting factor. It is worth being precise about what role it plays: it lowered the ceiling of the teacher signal, but it did not cause the transfer failure. A stronger APViT would have produced a stronger M-LDG, and possibly a marginally

better student, but not necessarily one that clears the published baseline. The capacity bottleneck remains regardless of this limitation.

The more productive direction for future work is probably not to keep refining the teacher, but to address the gap directly. Mirzadeh et al. [25] suggest that an intermediate-capacity assistant network can bridge large student-teacher size differences more effectively than direct distillation; that approach seems a natural fit here. Alternatively, techniques that sharpen or re-weight the soft label distributions before passing them to the student might help preserve more of the teacher signal across the gap. Either way, the lesson from this pipeline is that the bottleneck is structural, and architectural interventions that ignore it are unlikely to move the needle.

Conclusions and Future Work

“We can only see a short distance
ahead, but we can see plenty there
that needs to be done.”

Alan M. Turing

This chapter distills the main findings of this research, drawing on the experimental evidence reported in the Results chapter and the interpretive analysis developed in the previous chapter.

8.1. Conclusions

- The hypothesis is rejected: no enhanced EfficientFace configuration surpassed the published AffectNet baseline. The proposed variant, EfficientFace trained with soft labels from the M-LDG, reached 63% on AffectNet-7 and 55.99% on AffectNet-8 against a published reference of 63.70% and 59.89%.
- APViT guidance substantially improved the M-LDG, adding roughly 8 to 9 percentage points and lifting it to 64.34% on AffectNet-7 and 57.71% on AffectNet-8, confirming that the first hop of the two-stage transfer pipeline behaved as intended.
- The second hop of the transfer pipeline was not able to bridge the accuracy gap between the M-LDG and EfficientFace. We attribute this to the large capacity (in millions of parameters) gap between the two networks. With a ratio of approximately 44 to 1, the label distribution learning approach hits a bottleneck. This aligns with existing literature on knowledge distillation when using networks with large differentials [25].
- FER-pretrained backbone weights and local-global feature extraction modules both contributed negligibly to M-LDG accuracy in Phase I, though for different reasons: the former’s pretrained weights covered only the first three backbone stages, leaving the upper layers and classification head without FER-specific initialization, while the latter’s modules were designed for the shallow EfficientFace architecture and became redundant in the deeper IR50.

- Replacing the full APViT transformer with the lighter M-LDG as the direct teacher for EfficientFace in Phase II produced no statistically significant difference in student accuracy, suggesting that the computationally lighter intermediary can stand in for the heavier model without a meaningful loss in transfer quality; this finding remains tentative, however, given the limited number of experimental runs available for that comparison.

8.2. Future Work

- Reducing the M-LDG network size to better approximate the capacity of EfficientFace could improve the second hop of the transfer pipeline. Another approach could be to discard the APViT component and instead replicate the study with another network acting as intermediary. This keeps in line with the evidence from Mirzadeh et al.’s [25] where carefully selected intermediary network sizes can improve the transfer efficiency.
- The contempt class warrants deeper investigation: its recall remained near zero for the ablation baseline and recovered only partially under the optimized M-LDG, and the tentative observation that contempt recognition relies predominantly on the mouth region should be validated with quantitative, region-weighted evidence.
- Revisiting the pipeline with publicly available AffectNet-pretrained weights, should provide a better gain in performance. This could be with the same APViT network or choosing another similar method that is confirmed to have the weights available.
- The lack of statistical power in the phase II experiments warrants a larger sample size. This could be achieved by introducing additional factors into the experimental design like use of upscaling and teacher network size. This could

serve to better approximate the hard limit of the knowledge transfer pipeline as well as finding better configurations for the second hop. Additionally, prescinding from the already successful Phase I study could free up resources for additional replications in the second phase, increasing the statistical power of the study.

This section lists the public repositories and experiment-tracking resources used to implement and execute the experimental design.

9.1. Source Code

Main code repository: The primary repository for this thesis contains the modified LDG and EfficientFace implementations, dataset preprocessing scripts, and Jupyter notebooks for experiment execution and hyperparameter optimization. It is available at <https://github.com/DavidCrgh/FER-MLDG>.

Modified APViT codebase: Adaptations to the public APViT [36] implementation for training on AffectNet are available at <https://github.com/DavidCrgh/APViT>.

Modified EfficientFace codebase: A tweaked version of the EfficientFace [40] source code that fixes minor bugs and limitations of the original implementation is available at <https://github.com/DavidCrgh/EfficientFace>.

9.2. Dataset

The AffectNet dataset [27] used in this thesis can be requested from <https://mohammadmahoor.com/pages/databases/affectnet/>.

9.3. Gathered Data from the Experiments

Training metrics, validation curves, confusion matrices, network weights, and hyperparameter sweep results were logged with WandB. The project is available at <https://wandb.ai/dval/FER-Thesis-mLDG-Training>.

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